CORRELATED SAFETY ENVELOPES FOR DELAYED DUAL-PRICE CONTROL

HAOYI ZHANG*

Abstract. Inverse-price rate proposals need not remain capacity-feasible when sources use different versions of correlated advice. We study positive affine path prices over a latent box, distinguishing a shared realization, a bounded number of independently stamped realizations, and independent rowwise uncertainty. For prices with at most k uncertain coordinates and relative radius at most $\theta < 1$, a finite-dimensional polynomial optimization characterizes the sharp factor comparing the rowwise envelope with the r-stamp envelope. Coefficient symmetrization proves the universal inequality, and complete signed support families approach equality. We obtain explicit binomial formulas for one and two stamps, closed forms for two-coordinate prices with three, four, or five stamps, and an all-stamp support-two sandwich whose excess over one decays exponentially up to a square-root factor. The minimum stamp count attaining the rowwise envelope equals the chromatic number of a sign-conflict graph and can grow logarithmically in dimension even for two-coordinate prices. Exact saturation is decidable in polynomial time for one or two stamps and is NP-complete for every fixed stamp count at least three. Exact one-stamp safety certification is coNP-complete for two-coordinate prices of relative radius $1/4$; separator inequalities provide checkable upper certificates on structured inputs. Common attenuation and an acknowledged decrease-before-increase transition preserve commanded packing feasibility. A separate contraction argument gives sufficient conditions for convergence of the virtual dual iteration and vanishing safety conservatism. Exact finite checks include a correlation-aware command experiment with an unsafe unbarriered-mixing control; no packet-level or prediction-driven acceleration guarantee is claimed.

Key words. network utility maximization, robust feasibility, asynchronous optimization, correlated uncertainty, extremal inequalities, certification

MSC codes. 68Q25, 68W15, 90C25, 90C35

1. Introduction.

Price-based control decomposes global capacity constraints into local decisions. Under logarithmic utility, weight w_s proposes w_s/q_s at positive path price q_s. Equilibrium prices give optimal feasible allocations; intermediate, mispredicted, or mixed-version prices need not. Commanded-rate feasibility is therefore distinct from eventual convergence.

The difficulty is already visible with two sources sharing a link of capacity 3. Suppose their proposed rates are

$$x_1(z) = \frac{1}{1 + z/2}, \qquad x_2(z) = \frac{1}{1 - z/2}, \qquad -1 \le z \le 1. \tag{1.1}$$

For a common z, their sum is $2/(1 - z^2/4) \le 8/3$. Thus the entire correlated family is feasible with a margin of $1/3$. Nevertheless, a source using $z = -1$ and another using $z = 1$ each proposes rate 2. Their aggregate proposal is 4, not $8/3$. Every scalar price still lies in its prescribed interval. What failed was the shared-realization condition that justified adding the two rates.

Ignoring lost correlation retains a certificate the example refutes. Summing independently maximized source rates is safe but can overreserve capacity relative to every admissible correlated realization. We quantify this cost, its dependence on bounded version count, and exact certification of less conservative envelopes.

*Xi'an Jiaotong-Liverpool University, Suzhou, China (hyeliozhang@gmail.com).

1.1. The mathematical question. We consider a finite collection of positive reciprocal-affine functions

$$f_s(z) = \frac{v_s}{a_s + b_s^T z}, \qquad a_s > \|b_s\|_1, \quad v_s > 0, \quad z \in [-1,1]^d. \tag{1.2}$$

For one monitored link, v_s incorporates both the utility weight and the source's routing coefficient. Let M_1 be the maximum of $\sum_s f_s(z)$ over one shared realization. Let R be the sum of the individual maxima. If at most r versions are visible and the version used by each source is unknown, the appropriate envelope is

$$M_r = \max_{z^1,\dots,z^r \in [-1,1]^d} \sum_s \max_{j \in [r]} f_s(z^j). \tag{1.3}$$

The maximum includes the assignment of sources to versions. It is not an assertion that arbitrary independent versions are dynamically reachable by a particular feedback system. Rather, it is the robust envelope justified by an information contract that gives only a common box and a bound on the number of independently stamped realizations.

Ambient dimension d counts uncertain quantities, row-support bound k their maximum number per path price, and stamp count r measures correlation retained within versions. They are not interchangeable: small row support need not let few stamps realize every independent row maximum.

1.2. Results and proof architecture. The main quantitative result gives sharp constants $K_{k,1}(\theta)$ and $K_{k,2}(\theta)$ such that

$$R \le K_{k,j}(\theta) M_j \quad (j = 1, 2) \tag{1.4}$$

whenever each row has at most k nonzero coefficients and $\|b_s\|_1/a_s \le \theta < 1$. The constants are finite binomial sums, not outputs of an optimization oracle. For example,

$$\begin{aligned} K_{1,1}(\theta) &= 1 + \theta, & K_{1,2}(\theta) &= 1, \\ K_{2,1}(\theta) &= \frac{2(1+\theta)}{2-\theta^2}, & K_{2,2}(\theta) &= \frac{2}{2-\theta}. \end{aligned} \tag{1.5}$$

Both bounds are attained by an explicit family of 2^k prices. The one-stamp argument averages over cube corners and symmetrizes the coefficient magnitudes. The two-stamp argument instead averages over antipodal corner pairs. Proving sharpness of the latter bound requires comparing all pairs of corners, including those that are not antipodal. A coordinate-flip argument supplies that comparison.

For arbitrary fixed k, r, a finite-dimensional polynomial maximization over distributions of stamp columns gives the exact sharp uniform factor. The universal inequality uses coefficient symmetrization; complete families over all k-coordinate supports approach equality by a uniform sampling comparison. For support two, this reduces to a quadratic optimization. In addition to (1.5), we obtain $K_{2,3}(\theta) = 3/(3-\theta)$, $K_{2,4}(\theta) = 6/(6-\theta)$, and $K_{2,5}(\theta) = 8/(8-\theta)$. These constants are approached as dimension grows rather than asserted to be attained in every finite dimension. For every larger stamp count we give explicit two-sided support-two bounds; for fixed θ, their gap from one decreases between orders 2^{-r} and $\sqrt{r}\,2^{-r}$.

The stamp threshold has a different description. Join two sources when their affine prices have coefficients of opposite signs in some latent coordinate. We prove

that $M_r = R$ exactly when this sign-conflict graph can be colored with r colors. This yields a sharp complexity dichotomy: exact saturation is decidable in polynomial time for $r \le 2$ and is NP-complete for every fixed $r \ge 3$, even at relative radius at most $1/2$. We also bound the deficit $R - M_r$ by two weighted vertex-deletion problems. A family of two-coordinate prices turns the threshold into a binary covering-array problem: its threshold is $\Theta(\log d)$, although its one-stamp ratio is exactly $K_{2,1}(\theta)$. This separates the amount of conservatism from the number of versions needed to eliminate it completely.

Computing the coherent envelope is a third issue. One-coordinate prices separate by coordinate and admit a direct exact calculation. Two-coordinate prices already encode unweighted Max-Cut with an exact affine relation between objective values. The resulting safety decision is coNP-complete even when all denominators stay between 3 and 5. For structured instances we give separator-message certificates whose local inequalities imply a global upper bound. The dynamic-programming mechanism is standard; its role here is to make the particular robust envelope checkable without trusting its constructor.

Finally, we distinguish an envelope from its use in control. A largest safe common attenuation is obtained from the link envelopes. An acknowledged decrease-before-increase transition preserves packing feasibility between successive safe targets, even when coordinate applications are asynchronous. An exact three-stamp experiment exhibits the resulting correlation gain and an unsafe unbarriered-mixing negative control. Convergence of the virtual dual state is established only under a stated weighted diagonal-dominance condition, bounded read ages, bounded update gaps, and controlled additive errors. Exact convergence additionally requires vanishing errors; vanishing conservatism requires shrinking price uncertainty. None of the envelope inequalities by itself supplies those hypotheses.

1.3. Relation to earlier work. The inverse-price response comes from network utility maximization. Kelly et al.'s proportional-fairness model identifies prices as congestion signals [33]. Low and Lapsley analyze asynchronous gradient-projection flow control [38], Low, Peterson, and Wang characterize the Vegas equilibrium [39], and Low gives a broader dual model of TCP and queue management [37]. Decomposition and cross-layer formulations explain how local responses arise from global utility problems [43, 19]. These works characterize equilibria or trajectories; our object is the aggregate load of a fixed response family when a common realization is only partially preserved.

The closest safety-oriented pricing work addresses a complementary algorithmic question. The safe dual gradient method uses a diminishing margin and directional dual steps for linearly constrained NUM [51]. Later methods learn price responses while keeping every induced demand feasible under their stated assumptions, with high-probability safety in the bandit-feedback setting. They use a shrunk feasible set, bandit feedback, or gradual-variation assumptions and prove static or dynamic regret bounds [32, 52, 53]. Learning-NUM instead learns unknown utilities under queueing-delayed feedback and controls regret through parallel algorithm instances [24]. Those papers design prices or allocations under response uncertainty. We neither improve their regret guarantees nor infer an unknown response: we evaluate and certify the worst load of a declared reciprocal family under a bounded version-mixing contract. Conversely, stage-wise feasibility of one pricing algorithm does not supply the sharp M_r factors, stamp threshold, or certification complexity proved here.

Positive-linear-program and flow algorithms delimit a second boundary. Approx-

imation algorithms for packing and covering include parallel methods [40], fractional packing and covering schemes [45], and width-independent methods [55, 1]; multicommodity variants refine that paradigm [26, 13, 21]. Feasibility-preserving distributed packing [5], distributed maximum flow [27], iteration lower bounds [34], and specialized all-or-nothing or polymatroidal flow models [18, 17] concern algorithms that construct a packing or routing solution. No new packing solver is claimed here: the object is the worst load generated by a declared version-mixing contract.

Our delay analysis uses classical asynchronous results. Chaotic relaxation [16], asynchronous gradient methods [50], fixed-point theory [8], and the monograph treatment [10] establish convergence under explicit communication and update assumptions. Modern coordinate frameworks quantify related delay effects [44]; asynchronous consensus ADMM combines a partial barrier with a bounded-delay condition [56]. Recent primal–dual results cover totally asynchronous blocks, time-varying tracking, and bounded time-varying delays under their respective hypotheses [30, 49, 46]. Our contraction theorem is a sufficient specialization; convergence of a virtual price state does not certify capacity feasibility of every stamped command.

Robust optimization supplies the uncertainty language but not the same quantifiers. Robust convex optimization protects one decision against a common realization [7]; budgeted uncertainty organizes tractable counterparts and conservatism [12]. Finite adaptability and K-adaptability precompute a bounded policy menu and select a policy after a global scenario is observed [11, 28]. In M_r, by contrast, the adversary chooses, row by row, among r simultaneously available global stamps. This order of quantifiers is what creates the sign-conflict and column-distribution problems.

Learning-augmented algorithms seek consistency with accurate predictions while retaining robustness to bad advice. A foundational example is prediction-assisted caching [41]; the primal–dual framework of Bamas, Maggiori, and Svensson treats several online problems [6]. Robustness–consistency lower bounds and metric algorithms with untrusted predictions further develop that theory [54, 4]. Our guarantee is not competitive acceleration and uses no learned predictor. It is a deterministic feasibility reserve for a bounded reciprocal family; advice quality enters only through the certified uncertainty set and stamp contract.

Finally, the saturation threshold is a covering-array phenomenon. Universal test sets and qualitative independence were developed for exhaustive circuit testing [47]; the extremal column bound is due to Kleitman and Spencer [35]. Covering-array algorithms and surveys provide the broader combinatorial setting [29, 20], while maximum shattering uses a closely related column-frequency polynomial [2]. We import those extremal facts where stated. The reciprocal payoff, sharp reserve factors, conflict-graph equivalence, and exact safety-certification boundary are the claims proved here, rather than a renaming of the underlying covering or correlation inequalities [48].

2. Network prices and the information available to a certifier.

2.1. Packing constraints and logarithmic response. Let $A \in \mathbb{R}_+^{m\times n}$ be a fixed routing matrix and $c \in \mathbb{R}_{>0}^m$ a vector of link capacities. A commanded rate vector $x \in \mathbb{R}_+^n$ is feasible when

$$Ax \le c. \tag{2.1}$$

Every column of A is assumed nonzero. Thus each source uses some capacity-limited resource, and the feasible set is bounded. Binary routing matrices correspond to fixed paths; nonnegative coefficients also allow a fixed weighted use of resources. The state

is the commanded rate vector on the fixed routing matrix; route changes, queues, propagation of fluid already in a link, and packet scheduling require additional state.

For positive weights w_s, the utility maximization problem is

$$\max_{x>0} U(x) := \sum_{s=1}^{n} w_s \log x_s \quad \text{subject to } Ax \le c. \tag{2.2}$$

A sufficiently small strictly positive vector is strictly feasible. The feasible closure is compact. The objective tends to minus infinity if any coordinate tends to zero while the other coordinates remain bounded. Consequently an optimum exists in the strictly positive orthant. Strict concavity makes its rate vector x^* unique.

For a nonnegative link-price vector p, put $q = A^T p$. When every $q_s > 0$, the Lagrangian maximization separates:

$$\arg\max_{x_s>0}\{w_s \log x_s - q_s x_s\} = \frac{w_s}{q_s}. \tag{2.3}$$

Indeed the derivative is $w_s/x_s - q_s$, and the second derivative is $-w_s/x_s^2 < 0$. The dual function and its gradient are

$$\mathcal{D}(p) = c^T p + \sum_s w_s \left(\log \frac{w_s}{q_s} - 1\right), \tag{2.4}$$

$$\nabla \mathcal{D}(p) = c - A(w/q), \tag{2.5}$$

where division is coordinatewise. Slater's condition gives a dual optimum and the Karush–Kuhn–Tucker conditions. In particular, at a global dual optimum with positive path prices, the response in (2.3) is x^* and satisfies (2.1). A price vector away from that optimum need not generate a feasible response.

There is an important separation of variables here. The virtual dual iteration may evaluate w/q to compute (2.5); the commanded rate may be an attenuated version of that proposal. Replacing w/q by the commanded rate in the gradient defines a different iteration. Section 6 keeps the virtual calculation explicit so that its convergence proof is not inadvertently applied to a different feedback law.

2.2. Correlated positive affine prices. At a certification epoch we specify

$$q_s(z) = a_s + b_s^T z, \qquad z \in [-1,1]^d, \qquad a_s > \rho_s := \|b_s\|_1. \tag{2.6}$$

The center a_s, vector b_s, and weight w_s are fixed throughout this epoch. The strict margin makes every response well defined. For a monitored link ℓ, omit sources with $A_{\ell s} = 0$ and put $v_s = A_{\ell s} w_s$. Its load is

$$F_\ell(z) = \sum_{s:A_{\ell s}>0} \frac{v_s}{a_s + b_s^T z}. \tag{2.7}$$

Unless links must be compared, we suppress ℓ and call these terms rows. The sharp ratio statements concern a nonempty row set. A link used by no source has envelope zero and needs no ratio bound.

The same latent coordinate may increase one path price and decrease another. Such signed dependence arises, for example, from complementary adjustments to two component prices. It need not arise from every network model. If the uncertain variables are independent nonnegative increments of the actual link prices, then b_s

has no sign conflicts across sources, and all source maxima are compatible. In that case $M_1 = R$; exploiting correlation gives no improvement. Signed dependence is an assumption of the nontrivial examples, not a property supplied by the word "prediction."

An affine family can be induced by correlated link prices $p(z) = \bar{p} + Cz$ with $p(z) \geq 0$ and $a = A^T\bar{p}$, $b_s = C^T A_{:s}$. Our general envelope results only require (2.6). Where a lower-bound construction is used to establish a network phenomenon, we give a realization using nonnegative component prices and fixed paths. Thus the constructions do not rely on interpreting a negative affine coefficient as a negative physical link price.

The uncertainty set is deterministic. The averages over random signs in later proofs are an analytical device for lower-bounding a maximum. They do not impose a probability law on advice errors or turn an all-realization statement into a high-probability one. In particular, no calibration or independence of learned prediction errors is presumed.

2.3. One stamp, known cohorts, and unknown cohorts. With one common stamp, all sources use the same z, so the envelope is

$$M_1 = \max_{z\in[-1,1]^d} \sum_s f_s(z), \qquad f_s(z) = \frac{v_s}{a_s + b_s^T z}. \tag{2.8}$$

The independent rowwise envelope is explicit:

$$R = \sum_s m_s, \qquad m_s = \max_{z\in[-1,1]^d} f_s(z) = \frac{v_s}{a_s - \rho_s}. \tag{2.9}$$

For a known partition Π of the sources into stamped cohorts, different cohorts can use independent realizations, whereas sources within a cohort share a realization. Therefore

$$M(\Pi) = \sum_{C\in\Pi} \max_{z\in[-1,1]^d} \sum_{s\in C} f_s(z). \tag{2.10}$$

If only an upper bound r on the number of nonempty cohorts is known, then the certifier must consider every such partition. Equivalently it must consider (1.3). Repeated or unused stamps are allowed.

LEMMA 2.1 (Partition representation). *For every positive integer r,*

$$M_r = \max_{\Pi:\,|\Pi|\leq r} M(\Pi). \tag{2.11}$$

Moreover $M_1 \leq M_r \leq R$, the sequence M_r is nondecreasing, and $M_r = R$ for $r \geq n$.

Proof. For fixed copies $z^1, \ldots, z^r$, assign each row to an index attaining its largest response, breaking ties arbitrarily. The nonempty assignment classes form a partition into at most r cohorts. The sum of responses is at most $M(\Pi)$. Maximizing over the copies proves one direction of (2.11).

Conversely fix a partition with at most r cohorts. Continuity and compactness give a maximizing realization for every cohort. Use these realizations as the copies. A row's maximum over the copies is at least its response at the copy chosen for its own cohort. This proves the other direction. Additional copies can repeat existing ones, giving monotonicity. The upper bound follows by bounding each response by m_s. The partition into singleton rows attains R and uses at most n stamps. ∎

A second elementary bound is occasionally useful:

$$M_r \le rM_1. \tag{2.12}$$

For fixed copies, $\max_j f_s(z^j) \le \sum_j f_s(z^j)$ because all terms are positive. Sum over rows and then maximize. This bound depends only on the number of copies. It is different from the dimension-sensitive bounds on R/M_r proved below.

2.4. What a delay bound does and does not imply. In a slotted implementation with read age at most D, at most $D+1$ production times can occur in a window of retained advice. A certificate based only on that fact may use $r = D+1$. But the affine family in (2.6) must enclose every realization that the certificate allows for those stamps. If centers and radii change with production time, a box for the newest production time cannot simply be reused for earlier messages.

There are two valid choices. One may certify each actually known cohort against its own explicitly specified uncertainty set. Alternatively, one may construct a single common family enclosing all retained messages and use M_r for that family. The former can be tighter, but it is not the model of a common identical box analyzed by our sharp constants. The latter preserves the theorem's assumptions at the cost of possibly enlarging its radius.

Bounded delay alone does not make distinct stamps statistically independent. In this paper "independently stamped" means that no constraint relating their latent realizations is available to the certifier. A trajectory restriction such as $\|z^{j+1} - z^j\| \le \nu$ would reduce the uncertainty set. Our upper bounds remain safe after such a restriction, but their necessity and attaining constructions need not remain exact. This distinction is particularly important when using the chromatic threshold: it characterizes the stated information model, not all trajectories of every delayed control law.

2.5. Reduction to corners and finite encoding.

LEMMA 2.2 (Corner maximizers). *The maxima defining M_1, $M(\Pi)$, and M_r are attained at cube corners, using a corner for every independent copy.*

Proof. The reciprocal of a positive affine function is convex. More explicitly, its Hessian is $2v_s b_s b_s^T/(a_s + b_s^T z)^3$, which is positive semidefinite. A sum of these functions is convex. For fixed r, each function $f_s(z^j)$ is convex in the full vector of copies, so its maximum over j is also convex, as is the sum over s.

Every point in a cube is a convex combination of its corners. If G is convex and $z = \sum_v \lambda_v v$, then $G(z) \le \sum_v \lambda_v G(v) \le \max_v G(v)$. Applying this statement to a cohort, or to the product cube of all copies, proves the claim. Continuity ensures existence; no limiting argument at a zero denominator is involved. ∎

All computational statements use rational input coefficients written in binary and a rational capacity. A corner specifies only signs. Evaluating a response at such a corner uses rational additions and one division, and the positivity margin can be verified exactly. The numerator and denominator bit lengths of a sum of finitely many responses are bounded polynomially in the total input encoding length. This supplies short witnesses for unsafe instances despite the exponential number of corners. It does not make a complete enumeration polynomial.

2.6. Clipping advice without losing the certified dependence. The affine family may be used to constrain prediction advice, but the point at which clipping occurs matters. Suppose an advice source supplies an arbitrary vector $u \in \mathbb{R}^d$. A

command policy may first form $z = \mathrm{clip}_{[-1,1]^d}(u)$ and then compute every path price for that version as $a_s + b_s^T z$. This policy always produces a member of the certified family, even when the original advice is adversarial. It is a definition of the proposal policy, not an inference that an unobserved physical price lies inside the box.

Within one version, every source must use the same clipped latent vector, or a price value computed from that same vector. If different versions are retained, their number and assignment must obey the stamp contract. Once these conditions are met, positivity and the envelope inequality follow without any statistical assumption on u. A useful predictor may produce better proposals, but the safety statement does not require that benefit and does not establish it.

Clipping path prices independently is a different policy. It guarantees each price's marginal interval but can destroy the dependence used by M_1. In dimension two, consider the three prices and response weights

$$q_1 = 2 + z_1, \quad q_2 = 2 + z_2, \quad q_3 = 2 - (z_1 + z_2)/2, \qquad v = (1, 1, 4). \tag{2.13}$$

Each price lies in $[1, 3]$ on the certified box. At its four corners, the total reciprocal loads are $10/3$, $10/3$, $10/3$, and $14/3$, respectively, so $M_1 = 14/3$. Now take raw latent advice $u = (2, 0)$, whose unmodified prices are $(4, 2, 1)$. Independently clipping the three prices to $[1, 3]$ gives $(3, 2, 1)$ and total load

$$\frac{1}{3} + \frac{1}{2} + 4 = \frac{29}{6} > \frac{14}{3}.$$

The clipped vector has no representation (2.13) with one common z: its first two coordinates require $z_1 = 1, z_2 = 0$, which would make $q_3 = 3/2$, not one. At capacity $19/4$, the coherent certificate is safe but this independently clipped proposal is not.

In contrast, clipping the latent vector first gives $z = (1, 0)$, prices $(3, 2, 3/2)$, and load $7/2$. This proposal is inside the certified family. A policy that insists on clipping the prices independently must instead certify its actual image set or use the rowwise envelope. It cannot retain a coherent certificate merely because each individual price lies between its previously stated minimum and maximum.

There is an analogous distinction when the values are observations rather than policy outputs. Replacing an out-of-box observation by a clipped number does not prove that the actual observed state obeys the certificate. An implementation must either use the clipped value as the price governing its commanded proposal, or reject the old model and build a new enclosure for the observation. If a new proposal cannot be certified, retaining the previously feasible commanded allocation preserves the existing packing invariant. No increase justified only by a failed certificate is allowed.

This distinction gives a precise meaning to bounded advice in the present model. The bound applies to the value actually used to generate a proposal. A shared latent clipping rule preserves the dependencies that define a stamp; componentwise bounds alone do not. The argument is independent of how the advice was generated and therefore makes no claim about a trained predictor's accuracy, calibration, or operational reliability.

3. Sharp reserve factors. This section compares a sum of separately maximized rates with a jointly maximized load. The comparison is uniform over the number of sources, their positive weights, and the ambient dimension. Throughout, a nonempty family satisfies

$$|\operatorname{supp}(b_s)| \le k, \qquad \frac{\|b_s\|_1}{a_s} \le \theta < 1, \qquad k \ge 1, \quad 0 \le \theta. \tag{3.1}$$

The sharp constant is the smallest K for which $R \le K M_r$ holds for every finite family in this class. Sharpness may mean attainment by a finite family or approach by a sequence of finite families; we distinguish these cases.

3.1. Closed forms for one and two stamps. For $t \in [0,1)$ define

$$h_{k,1}(t) = 2^{-k} \sum_{j=0}^{k} \binom{k}{j} \frac{1}{1 + t(2j-k)/k}, \tag{3.2}$$

$$h_{k,2}(t) = 2^{-k} \sum_{j=0}^{k} \binom{k}{j} \frac{1}{1 - t|2j-k|/k}, \tag{3.3}$$

$$K_{k,j}(\theta) = \frac{1}{(1-\theta) h_{k,j}(\theta)} \quad (j = 1, 2). \tag{3.4}$$

Both sums have only $k+1$ terms. Their arguments record the sum of k independent signs. The second sum uses the better of a sign vector and its negative.

THEOREM 3.1 (Sharp one- and two-stamp bounds). *Every family satisfying* (3.1) *obeys*

$$R \le K_{k,1}(\theta) M_1, \qquad R \le K_{k,2}(\theta) M_2. \tag{3.5}$$

For each fixed k and $0 \le \theta < 1$, both constants are best possible and are attained by the family

$$f_\sigma(z) = \frac{1}{1 + (\theta/k)\sigma^T z}, \qquad \sigma \in \{-1, 1\}^k. \tag{3.6}$$

The bounds need not coincide. For two-coordinate prices, the one-stamp factor is $2(1+\theta)/(2-\theta^2)$, whereas the two-stamp factor is $2/(2-\theta)$. Merely doubling the number of independent versions does not generally eliminate the cost of discarding correlation.

LEMMA 3.2 (Coefficient symmetrization). *Let $c \in \mathbb{R}^k_+$ have sum $t < 1$, and let ε be uniform on $\{-1,1\}^k$. Then*

$$\mathbb{E} \frac{1}{1 + c^T \varepsilon} \ge h_{k,1}(t), \tag{3.7}$$

$$\mathbb{E} \frac{1}{1 - |c^T \varepsilon|} \ge h_{k,2}(t). \tag{3.8}$$

For $j = 1, 2$, the function $(1-t)h_{k,j}(t)$ is nonincreasing on $[0,1)$.

Proof. On the open unit ℓ_1 ball, the function $c \mapsto (1 + c^T\varepsilon)^{-1}$ is convex. Its expectation is convex and invariant under permutations of the coordinates. Average the $k!$ permutations of c. Their mean is $(t/k, \dots, t/k)$, so Jensen's inequality gives (3.7).

For the second statement use

$$\frac{1}{1 - |c^T\varepsilon|} = \max\left\{ \frac{1}{1 + c^T\varepsilon}, \frac{1}{1 - c^T\varepsilon} \right\}.$$

A maximum of convex functions is convex. Averaging over signs again makes the function permutation invariant, so the same averaging argument proves (3.8). This reasoning includes zero coordinates; padding a shorter row with zeros is legitimate.

Each summand of $(1-t)h_{k,1}(t)$ has the form $(1-t)/(1+tZ)$, with $Z \in [-1,1]$. Its derivative is $-(1+Z)/(1+tZ)^2 \le 0$. For the second sum use $Z = -|2j-k|/k$, which lies in the same interval. The asserted monotonicity follows by averaging these derivatives. ∎

Proof of the inequalities in Theorem 3.1. Normalize a row by a_s. Its coefficient magnitudes, padded to length k, sum to $t_s = \|b_s\|_1/a_s \le \theta$. A uniform global corner $\varepsilon \in \{-1,1\}^d$ induces uniform independent signs on the row's support, irrespective of the signs of b_s. Hence

$$\mathbb{E} f_s(\varepsilon) \ge \frac{v_s}{a_s} h_{k,1}(t_s).$$

Since $m_s = v_s/[a_s(1-t_s)]$, Lemma 3.2 gives

$$m_s \le K_{k,1}(\theta)\frac{v_s}{a_s} h_{k,1}(t_s) \le K_{k,1}(\theta)\mathbb{E} f_s(\varepsilon).$$

The same global random corner is used for every row. Summing and applying $\mathbb{E}\sum_s f_s(\varepsilon) \le M_1$ proves the first inequality. No independence between different rows was used.

For two stamps use the single global pair $(\varepsilon, -\varepsilon)$. A row's better response is

$$\max\{f_s(\varepsilon), f_s(-\varepsilon)\} = \frac{v_s/a_s}{1 - |b_s^T\varepsilon|/a_s}.$$

Apply (3.8) and the same monotonicity argument row by row. The expected sum of these better responses is at most M_2, because every realized antipodal pair is admissible in its definition. This proves the second inequality. ∎

The common random experiment is essential. Independently choosing a favorable pair for each row would prove a statement about R, not a lower bound on M_2.

LEMMA 3.3 (An extremal antipodal pair). *For the sign family* (3.6)*, a pair of antipodal corners maximizes the two-stamp load among all pairs of corners.*

Proof. By coordinate sign changes, one corner can be taken to be the all-positive corner. Let the second differ from it in h coordinates. Averaging over the row index σ, let T be the sum of the h signs in differing coordinates and let U be the sum of the signs in the remaining common coordinates. The smaller of the two dot products is $U - |T|$. Therefore the average row response is the expectation of

$$\phi(U-|T|), \qquad \phi(u) = \frac{1}{1+(\theta/k)u}.$$

The function ϕ is nonincreasing on every argument that occurs.

Suppose one common coordinate remains. Separate its sign $\epsilon \in \{-1,1\}$ from U, writing the other common signs as U'. Before this coordinate is changed into a differing coordinate, the argument is $U' + \epsilon - |T|$. Afterwards it is $U' - |T+\epsilon|$. Condition on T and U'. If $|T| \ge 1$, the two arguments, as multisets over the two values of ϵ, are identical. If $T = 0$, the new argument is always $U'-1$, whereas the old arguments are $U'-1$ and $U'+1$. In the latter case the new conditional expectation is no smaller. These cases are exhaustive because T is an integer sum of signs.

Thus increasing the Hamming distance between the two corners never decreases the averaged response. Repeating the operation makes them antipodal. Multiplication by the number 2^k of rows proves the result. ∎

Proof of sharpness in Theorem 3.1. Every row of (3.6) has maximum $(1-\theta)^{-1}$, so $R = 2^k/(1-\theta)$. For any fixed corner z, the multiset of products $\sigma_i z_i$ is uniform over all sign vectors as σ varies. Consequently the total load is $2^k h_{k,1}(\theta)$ at every corner. Lemma 2.2 gives $M_1 = 2^k h_{k,1}(\theta)$.

For an antipodal pair, the same counting gives total load $2^k h_{k,2}(\theta)$. Lemmas 2.2 and 3.3 show that no non-corner or non-antipodal pair improves this value. Dividing R by these two exact maxima proves both sharpness statements. When $\theta = 0$ all rows are constant and both factors equal one. ∎

3.2. Dependence on support and radius. The binomial sums reveal two distinct limiting regimes. For fixed k,

$$\lim_{\theta\uparrow 1} K_{k,1}(\theta) = 2^k, \qquad \lim_{\theta\uparrow 1} K_{k,2}(\theta) = 2^{k-1}. \tag{3.9}$$

Indeed, after multiplication by $1-\theta$, only the all-negative sign outcome remains in (3.2); in (3.3), both constant sign outcomes remain. Their probabilities are 2^{-k} and 2^{1-k}, respectively. Every other term has a denominator bounded away from zero in this limit.

For fixed $\theta < 1$, the support dependence is much milder:

$$\lim_{k\to\infty} K_{k,j}(\theta) = \frac{1}{1-\theta} \quad (j = 1, 2). \tag{3.10}$$

To see this quantitatively, write $Z = k^{-1}\sum_{i=1}^{k} \varepsilon_i$, so $\mathbb{E}Z^2 = 1/k$. Symmetry yields $h_{k,1}(\theta) = \mathbb{E}(1-\theta^2 Z^2)^{-1}$. Hence

$$1 + \frac{\theta^2}{k} \le h_{k,1}(\theta) \le 1 + \frac{\theta^2}{k(1-\theta^2)}. \tag{3.11}$$

The lower bound uses $(1-u)^{-1} \ge 1+u$; the upper bound uses $u/(1-u) \le u/(1-\theta^2)$ for $0 \le u \le \theta^2$. Similarly, Cauchy–Schwarz gives

$$1 \le h_{k,2}(\theta) \le 1 + \frac{\theta}{(1-\theta)\sqrt{k}}. \tag{3.12}$$

The two limits (3.9) and (3.10) therefore cannot be interchanged without specifying a joint scaling. A bound that keeps only 2^k can be very pessimistic when the relative radius stays separated from one.

Every $h_{k,j}(\theta)$ is at least one, and retaining just the singular terms in its binomial sum gives

$$K_{k,1}(\theta) \le \min\{2^k, (1-\theta)^{-1}\}, \quad K_{k,2}(\theta) \le \min\{2^{k-1}, (1-\theta)^{-1}\}. \tag{3.13}$$

The factors are nondecreasing in θ by Lemma 3.2. These properties also provide inexpensive consistency checks for their exact evaluation.

3.3. A finite variational formula for any stamp count. There is a sharp characterization for general r, although its finite optimization can be large. The use of column distributions parallels the shattering Lagrangian in [2]; here the payoff is a reciprocal response rather than a complete-coverage indicator. A column of an r-stamp corner matrix is a sign vector in $\{-1, 1\}^r$. Complementing a column does not change an average over all coefficient signs. We therefore use the alphabet

$$\mathcal{C}_r = \{u \in \{-1, 1\}^r : u_1 = 1\}, \qquad |\mathcal{C}_r| = 2^{r-1}. \tag{3.14}$$

It includes the constant column. For k columns define

$$\Phi_{k,r,t}(u^1,\dots,u^k) = 2^{-k} \sum_{\varepsilon\in\{-1,1\}^k} \frac{1}{1+(t/k)\min_{j\in[r]}\sum_{i=1}^k \varepsilon_i u_j^i}. \tag{3.15}$$

If π is a probability distribution on $\mathcal{C}_r$, let

$$G_{k,r}(\pi;t) = \sum_{u^1,\dots,u^k\in\mathcal{C}_r} \pi_{u^1}\cdots\pi_{u^k}\Phi_{k,r,t}(u^1,\dots,u^k), \qquad G^*_{k,r}(t) = \max_\pi G_{k,r}(\pi;t). \tag{3.16}$$

For fixed t, this is a degree-k polynomial on a compact simplex, so its maximum exists. This description is finite, but the alphabet is exponential in r and the direct expression has $2^{k(r-1)}$ terms. It is not a polynomial-time prescription for unrestricted k, r.

THEOREM 3.4 (Sharp factor for arbitrary stamps). *For every positive k, r and $0 \le \theta < 1$, the sharp uniform factor over the class* (3.1) *is*

$$K_{k,r}(\theta) = \frac{1}{(1-\theta)G^*_{k,r}(\theta)}. \tag{3.17}$$

For $r = 1, 2$ this agrees with (3.4)*. For $\theta > 0$ and general r, sharpness is approached by families having $2^k\binom{d}{k}$ rows, each supported on exactly k coordinates, as $d \to \infty$ with k, r, θ fixed. At $\theta = 0$ every row is constant and the factor 1 is attained trivially.*

Universal inequality. Choose a distribution π attaining $G^*_{k,r}(\theta)$. For every latent coordinate independently, draw a column from π and then complement it with probability $1/2$. The resulting $r \times d$ matrix specifies r admissible global corners.

Normalize a row by a_s, replace its coefficient signs by the corresponding independent column complements, adjoin dummy zero coordinates if necessary, and pad its magnitude vector with zeros to length k. For a nonnegative magnitude vector c with $\sum_i c_i = t < 1$, the expected best response, without its multiplier v_s/a_s, is

$$g_\pi(c) = \mathbb{E}\max_{j\in[r]} \frac{1}{1+\sum_{i=1}^k c_i\epsilon_i u_j^i}.$$

This is a convex function of c: every reciprocal term is convex on the domain, and a maximum and an expectation preserve convexity. Since the columns are independent and identically distributed, it is also invariant under coordinate permutations. Averaging permutations as before gives

$$g_\pi(c) \ge g_\pi(t/k,\dots,t/k) = G_{k,r}(\pi;t).$$

For each outcome of the columns and their signs, the balanced-row response is $(1+tZ)^{-1}$ for some $Z \in [-1,1]$. Thus $(1-t)G_{k,r}(\pi;t)$ is nonincreasing by the derivative calculation used in Lemma 3.2. In particular,

$$(1-t_s)\mathbb{E}\max_j f_s(z^j) \ge \frac{v_s}{a_s}(1-\theta)G^*_{k,r}(\theta).$$

After dividing by $1-t_s$, the right side is $m_s(1-\theta)G^*_{k,r}(\theta)$. Sum over rows. The expected sum for the single jointly generated stamp matrix is at most M_r. Rearranging proves the universal inequality with (3.17). ∎

To prove sharpness we need a family that exposes a single distribution of stamp columns to all possible small supports. Fix $d \geq k$. For every k-element subset $S \subseteq [d]$ and every sign assignment $\sigma \in \{-1, 1\}^S$, include

$$f_{S,\sigma}(z) = \frac{1}{1 + (\theta/k) \sum_{i \in S} \sigma_i z_i}. \tag{3.18}$$

There are $N_d = 2^k \binom{d}{k}$ rows and $R = N_d/(1-\theta)$. This construction averages not only over signs, but also over all supports of size k.

LEMMA 3.5 (Sampling columns with and without replacement). *For a fixed stamp matrix, normalize every column into $\mathcal{C}_r$ and let π_d be its empirical distribution. Let H_d be its load for* (3.18)*, divided by N_d. Then*

$$|H_d - G_{k,r}(\pi_d; \theta)| \leq \frac{k(k-1)}{2d} \left(\frac{1}{1-\theta} - \frac{1}{1+\theta} \right). \tag{3.19}$$

Proof. The value H_d is the average of (3.15) for k uniformly chosen distinct columns. Because Φ is symmetric in its arguments, it does not matter whether the distinct columns are ordered. The value $G_{k,r}(\pi_d; \theta)$ averages the same expression when the k column indices are drawn independently with replacement. Condition this latter sampling on all indices being distinct. Its conditional average is exactly H_d.

The probability of a repeated index is at most $\binom{k}{2}/d$, by a union bound over pairs of sampled positions. Every response in the average lies between $(1+\theta)^{-1}$ and $(1-\theta)^{-1}$. Decomposing the unconditional expectation according to whether a repetition occurs proves (3.19). ∎

Sharpness in Theorem 3.4. Lemma 3.5 holds for every stamp matrix, so

$$\frac{M_r}{N_d} \leq G^*_{k,r}(\theta) + O(d^{-1}),$$

where the displayed constant in (3.19) is uniform over all matrices. Conversely, empirical distributions with denominator d approximate every point of the fixed finite simplex as $d \to \infty$. Choose π_d approaching a maximizer π^* and realize its integer column counts by a stamp matrix. Polynomial continuity gives $G_{k,r}(\pi_d; \theta) \to G^*_{k,r}(\theta)$. The same lemma shows that this matrix's normalized load has the same limit. Therefore

$$\lim_{d \to \infty} \frac{M_r}{N_d} = G^*_{k,r}(\theta), \qquad \lim_{d \to \infty} \frac{R}{M_r} = \frac{1}{(1-\theta) G^*_{k,r}(\theta)}.$$

Together with the universal inequality this proves sharpness. For $r = 1, 2$, Theorem 3.1 has already identified the unique sharp constant, so the two expressions necessarily agree. Equivalently, for one stamp the alphabet has one column; for two stamps Lemma 3.3 shows that concentrating on the alternating column maximizes the average. ∎

The theorem does not require a rational optimizer. For rational θ, rational empirical distributions and prices approximate a maximizing distribution, which suffices for sharpness.

3.4. Two-coordinate prices and a quadratic palette problem. For $k = 2$ and $r \geq 2$, the general variational formula simplifies. Let $\mathcal{V}_r$ be the nonconstant

columns in $\mathcal{C}_r$, so $|\mathcal{V}_r| = 2^{r-1} - 1$. For $u, v \in \mathcal{V}_r$, define

$$N(u,v) = |\{(u_j, v_j) : j \in [r]\}| \in \{2,3,4\}, \quad \beta_r = \frac{1}{4} \max_{\pi \in \Delta(\mathcal{V}_r)} \sum_{u,v} \pi_u \pi_v N(u,v). \tag{3.20}$$

Here $N(u,u) = 2$. The quantity counts how many of the four sign pairs can be selected by the stamps. It depends only on r, not on the price radius.

THEOREM 3.6 (Sharp support-two factors). *For every $r \geq 2$,*

$$K_{2,r}(\theta) = \frac{1}{1 - (1-\beta_r)\theta}. \tag{3.21}$$

In particular,

$$K_{2,2}(\theta) = \frac{2}{2-\theta}, \qquad K_{2,3}(\theta) = \frac{3}{3-\theta}, \qquad K_{2,4}(\theta) = \frac{6}{6-\theta}, \qquad K_{2,5}(\theta) = \frac{8}{8-\theta}. \tag{3.22}$$

The two-stamp value is attained by (3.6)*. The three-, four-, and five-stamp values are sharp as limits of* (3.18) *with $k = 2$.*

Reduction to the quadratic problem. Consider the four sign rows on two coordinates with equal coefficient magnitude $\theta/2$. If both stamp columns are nonconstant, each coordinate takes both signs. For any of the four desired minimizing sign pairs, its row reaches response $(1-\theta)^{-1}$ exactly when that pair occurs among the stamps. If the pair does not occur, a pair at Hamming distance one does occur: otherwise the only available pair would be its opposite, making both columns constant. The best response for a missing pair is therefore 1. Averaging over the four row signs gives

$$\Phi_{2,r,\theta}(u,v) = \frac{N(u,v)}{4(1-\theta)} + 1 - \frac{N(u,v)}{4}. \tag{3.23}$$

A constant column can be replaced by any nonconstant column without reducing the summed response of these four rows. If the other column is nonconstant, the old pair has two adjacent patterns and total response $2/(1-\theta) + 2$; the new pair has at least two patterns and is no worse by (3.23). If the other column is constant, the old total is $(1-\theta)^{-1} + 2 + (1+\theta)^{-1}$ and the new total is $2/(1-\theta) + 2$, again no smaller. Replacing all occurrences of the constant alphabet symbol by one fixed nonconstant symbol thus cannot reduce $G_{2,r}(\pi;\theta)$. An optimizer may be restricted to $\mathcal{V}_r$.

Equation (3.23) now gives

$$G^*_{2,r}(\theta) = \frac{\beta_r}{1-\theta} + 1 - \beta_r.$$

Substitution into Theorem 3.4 proves (3.21), including its sharpness statement. ∎

Evaluation for two through five stamps. For $r = 2$ there is only one nonconstant column modulo complement. Its self-pair has $N = 2$, so $\beta_2 = 1/2$.

For $r = 3$ there are three nonconstant column classes. Two distinct classes have $N = 3$, and a self-pair has $N = 2$. Consequently

$$\sum_{u,v} \pi_u \pi_v N(u,v) = 3 - \sum_u \pi_u^2 \leq \frac{8}{3}.$$

Equality holds at the uniform distribution on the three classes, yielding $\beta_3 = 2/3$.

For $r = 4$ split the seven classes into three balanced classes, having two occurrences of each sign, and four unbalanced classes. Distinct balanced classes have $N = 4$. Every other pair of distinct classes has $N = 3$, and diagonal pairs have $N = 2$. Let b be the total probability assigned to balanced classes. The objective is

$$3 + b^2 - 2 \sum_{u \text{ balanced}} \pi_u^2 - \sum_{u \text{ unbalanced}} \pi_u^2.$$

For fixed b, Cauchy–Schwarz bounds this by

$$3 + b^2 - \frac{2b^2}{3} - \frac{(1-b)^2}{4} = \frac{11}{4} + \frac{b}{2} + \frac{b^2}{12} \le \frac{10}{3}.$$

The last polynomial is increasing on $[0, 1]$. Equality holds when $b = 1$ and the three balanced classes are equally likely. Hence $\beta_4 = 5/6$.

For $r = 5$, identify a canonical nonconstant column with a nontrivial split of a five-element set, modulo complement. There are five singleton splits and ten two-splits. Distinct splits give three patterns except when two two-splits cross, in which case they give four; a diagonal gives two. Let $x \in \mathbb{R}_+^5$ and $y \in \mathbb{R}_+^{10}$ be their probability vectors, let $b = \mathbf{1}^T y$, and let A be the adjacency matrix of the line graph $L(K_5)$ on the two-splits. Then

$$\mathbb{E}N = 3 - \|x\|_2^2 - \|y\|_2^2 + y^T A y.$$

If B is the unsigned vertex–edge incidence matrix of K_5, then $A = B^T B - 2I$ and $BB^T = 3I + J$. Thus A has eigenvalues $6, 1, -2$ with multiplicities $1, 4, 5$. Writing $y = (b/10)\mathbf{1} + z$, $z \perp \mathbf{1}$, gives

$$y^T(A - I)y = \frac{b^2}{2} + z^T(A - I)z \le \frac{b^2}{2}.$$

Consequently $\mathbb{E}N \le 3 + b^2/2 \le 7/2$. Equality holds when $x = 0$ and y is uniform on the ten two-splits (also on any four-edge star), so $\beta_5 = 7/8$. Substitution in (3.21) proves (3.22). ∎

For all finite r, $1/2 \le \beta_r < 1$. The lower bound follows by using one nonconstant column. For the strict upper bound, put $L = |\mathcal{V}_r|$. Since diagonal pairs have value two and all others have value at most four,

$$\beta_r \le 1 - \tfrac{1}{2} \sum_u \pi_u^2 \le 1 - \frac{1}{2L} < 1.$$

Thus a bounded number of stamps leaves some worst-case correlation benefit for positive radius, even at support two. The factor is nonincreasing in r, because M_r is nondecreasing for every family and the factor is its sharp supremum ratio. This monotonicity does not require solving the quadratic problem.

3.5. Finite extremizers approaching the support-two constants. The complete-pair family has four rows for every unordered coordinate pair. Write $C = \binom{d}{2}$ and $T_3(d) = \lfloor d^2/3 \rfloor$. For $r = 2, 3, 4, 5$, the exact maximum number of covered pair patterns, counted over coordinate pairs, is

$$\begin{aligned} &P_2(d) = 2C, && P_3(d) = 2C + T_3(d), \\ &P_4(d) = 2C + 2T_3(d), && P_5(d) = \begin{cases} 4, & d = 2, \\ \lfloor 7d^2/4 - d \rfloor, & d \ge 3. \end{cases} \end{aligned} \tag{3.24}$$

It follows that

$$R = \frac{4C}{1-\theta}, \qquad M_r = 4C + \frac{\theta}{1-\theta} P_r(d). \tag{3.25}$$

These identities make the asymptotic sharpness concrete, without numerically optimizing the palette polynomial.

Proof of (3.24). As in the preceding proof, constant columns may be removed without decreasing the load. For two stamps there is one remaining class, and every coordinate pair contributes two patterns. For three stamps, coordinates in different classes contribute three patterns and coordinates in the same class contribute two. If the three class sizes are n_1, n_2, n_3, the number of cross-class pairs is $(d^2 - \sum_i n_i^2)/2$. This is maximized when the sizes differ by at most one, giving $T_3(d)$.

For four stamps, let b_1, b_2, b_3 be the sizes of the balanced classes and let $u_1, \dots, u_4$ be the sizes of the unbalanced classes. Put $n_B = \sum_j b_j$, $n_U = \sum_i u_i$, so $n_B + n_U = d$. Relative to the baseline of two patterns per coordinate pair, every pair from two distinct classes contributes one extra pattern, and a pair from two distinct balanced classes contributes one further pattern. Hence the total extra contribution is

$$E = \frac{d^2 - \sum_i u_i^2 - \sum_j b_j^2}{2} + \frac{n_B^2 - \sum_j b_j^2}{2}.$$

Cauchy–Schwarz gives $\sum_i u_i^2 \ge n_U^2/4$ and $\sum_j b_j^2 \ge n_B^2/3$. Therefore

$$E \le \frac{1}{2}\left(d^2 - \frac{(d-n_B)^2}{4} + \frac{n_B^2}{3} \right).$$

Its derivative with respect to n_B is $(d-n_B)/4 + n_B/3$, which is nonnegative on $[0, d]$. The right-hand side is therefore at most its value $2d^2/3$ at $n_B = d$. Since E is an integer, $E \le \lfloor 2d^2/3 \rfloor = 2\lfloor d^2/3 \rfloor$. Equality is obtained by using only the three balanced classes with sizes differing by at most one. This proves the four-stamp expression, including the case in which no balanced class is initially present.

For five stamps use the split representation from the preceding proof. If column-class counts are assembled into vectors x, y of total mass d, the ordered pattern sum is $3d^2 - \|x\|_2^2 + y^T(A-I)y \le 7d^2/2$. Since the d diagonal ordered pairs contribute two each, $P_5(d) \le \lfloor 7d^2/4 - d \rfloor$. For residues $d \equiv 0, 1, 3 \pmod 4$, distribute the columns as evenly as possible among the four two-splits incident to a fixed vertex; the resulting value $2d^2 - d - \sum_i n_i^2$ reaches the floor. If $d = 4q + 2 \ge 6$, take one copy of each of the six edges on vertices $1, 2, 3, 4$ and $q - 1$ further copies of each edge incident to vertex 1 in K_5. Its count vector is $B^T a$ for $a = (q - 1/2, 1/2, 1/2, 1/2, -1/2)$, so it lies in the 6- and 1-eigenspaces and attains the spectral upper bound. For $d = 2$, one pair realizes at most four patterns and two crossing classes attain four. Finally (3.23), summed over pairs, proves (3.25) for all four stamp counts. ∎

For instance, at $\theta = 1/2$ the sharp factors for three, four, and five stamps are $6/5$, $12/11$, and $16/15$. The complete-pair family in dimension four has ratios $48/41$, $24/23$, and 1, respectively. These are below the sharp limiting values, as they must be. A finite check that incorrectly required equality in every dimension would reject a correct asymptotic sharpness statement.

3.6. Certified finite-grid intervals. For rational θ, the variational value can be enclosed by exact enumeration. Let $m = 2^{r-1}$ and let L_Q be the maximum of

$G_{k,r}(\pi;\theta)$ over distributions whose masses are multiples of $1/Q$. Coupling two column draws and then k independent positions gives

$$|G(\pi) - G(\pi')| \le \frac{k}{2}\left(\frac{1}{1-\theta} - \frac{1}{1+\theta}\right)\|\pi - \pi'\|_1.$$

Rounding $m-1$ coordinates of a maximizer down and assigning the residual to the last coordinate therefore yields the certified interval

$$\begin{aligned} L_Q \le G^*_{k,r}(\theta) \le U_Q &:= \min\left\{\frac{1}{1-\theta}, L_Q + \frac{k(m-1)}{Q}\left(\frac{1}{1-\theta} - \frac{1}{1+\theta}\right)\right\}, \\ \frac{1}{(1-\theta)U_Q} \le K_{k,r}(\theta) &\le \frac{1}{(1-\theta)L_Q}. \end{aligned} \tag{3.26}$$

All endpoints are rational, $L_Q \ge 1$, and the grid has $\binom{Q+m-1}{m-1}$ points. This is a fixed-parameter certificate, not a uniform approximation scheme; a candidate distribution alone proves a sound factor but not sharpness.

3.7. Finite stamps cannot remove all uniform conservatism. For support one, two stamps eliminate the gap. For support at least two, no fixed finite number of stamps has that property uniformly over dimension. The strict separation follows from collisions between column types, even when the relative radius stays far below one.

PROPOSITION 3.7 (A quantitative noncollapse bound). *For $k \ge 2$, $r \ge 1$, and $0 < \theta < 1$,*

$$K_{k,r}(\theta) \ge \left(1 - \frac{\theta}{2^{r-1}\big(k(1-\theta) + 2\theta\big)}\right)^{-1} > 1. \tag{3.27}$$

Proof. Draw the first two canonical columns independently from a distribution π. They coincide with probability $\sum_u \pi_u^2$. Independently, their coefficient signs in (3.15) are opposite with probability $1/2$. On this event their contributions cancel at every stamp. The smallest possible dot product of the remaining $k-2$ signed columns is $-(k-2)$, so the reciprocal response is at most

$$\frac{1}{1 - \theta + 2\theta/k}.$$

On every other event it is at most $(1-\theta)^{-1}$. After multiplying by $1-\theta$, the expected deficit from one is therefore at least

$$\frac{1}{2}\left(\sum_u \pi_u^2\right)\left(1 - \frac{1-\theta}{1-\theta+2\theta/k}\right) = \left(\sum_u \pi_u^2\right)\frac{\theta}{k(1-\theta)+2\theta}.$$

Cauchy–Schwarz gives $\sum_u \pi_u^2 \ge 1/m = 2^{1-r}$. The inequality holds for every distribution, including a maximizer. Substitute it in (3.17) to obtain (3.27). The subtracted fraction is positive and less than one, proving strictness. ∎

The proposition concerns the sharp constant over all finite families. For a particular family, $r \ge n$ always attains R, and much smaller r can suffice. There is no contradiction: the extremal families used in Theorem 3.4 increase in size and dimension while r stays fixed. Equation (3.27) also quantifies only a lower bound on the worst ratio; it does not replace the sharp formulas where those have been evaluated.

COROLLARY 3.8 (All-stamp support-two sandwich). *For $r \geq 4$, put*

$$q_r = \binom{r-1}{\lfloor r/2 \rfloor - 1}.$$

Then, for $0 < \theta < 1$,

$$\frac{1}{1-\theta/2^r} \leq K_{2,r}(\theta) \leq \frac{1}{1-\theta/(2q_r)}. \tag{3.28}$$

Consequently, for fixed θ, $K_{2,r}(\theta) - 1$ is between order 2^{-r} and $O(\sqrt{r}\,2^{-r})$.

Proof. The lower bound is Proposition 3.7 with $k = 2$. For the upper bound, fix the first coordinate of each canonical column to $+1$. For every t-subset $S \subseteq \{2, \ldots, r\}$, $t = \lceil r/2 \rceil$, take the column negative exactly on S; there are q_r columns. Two distinct sets intersect because $2t > r - 1$ and, having equal size, have both differences nonempty. Their columns therefore realize $(+,+)$, $(-,-)$, and both mixed pairs, so $N = 4$ in (3.20); a diagonal pair has $N = 2$. The uniform distribution gives

$$\tfrac{1}{4}\mathbb{E}N = \tfrac{1}{4}[2/q_r + 4(1 - 1/q_r)] = 1 - (2q_r)^{-1}.$$

Theorem 3.6 proves the upper bound. The central-binomial estimate $q_r = \Theta(2^r/\sqrt{r})$ yields the orders. This is the qualitative-independence family behind the classical extremal bound [35]; only its reciprocal-payoff consequence is new here. For $r = 4, 5$, the construction meets the exact values above, so the upper side of (3.28) is an equality. ∎

3.8. The positive-price boundary as a pattern-coverage payoff. A second interpretation of the variational formula emerges as $\theta \uparrow 1$. For a canonical column tuple $u^1, \ldots, u^k$, let $N_k(u^1, \ldots, u^k)$ be the number of distinct sign patterns among its r rows, and set

$$\beta_{k,r} = \max_{\pi} \mathbb{E}_{u^1,\ldots,u^k \sim \pi} \frac{N_k(u^1, \ldots, u^k)}{2^k}. \tag{3.29}$$

Unlike the indicator that all 2^k patterns appear, this payoff gives credit for partial coverage. The distinction is relevant when $r < 2^k$, where full shattering is impossible but reciprocal loads still differ across stamp matrices.

PROPOSITION 3.9 (Boundary limit of the uniform factor). *For fixed positive k, r,*

$$\lim_{\theta \uparrow 1} K_{k,r}(\theta) = \beta_{k,r}^{-1}. \tag{3.30}$$

Furthermore,

$$1 - (1 - 2^{-k})^r \leq \beta_{k,r} \leq \min\{1, r/2^k\}. \tag{3.31}$$

For $k \geq 2$, the additional upper bound $\beta_{k,r} \leq 1 - 2^{-r} < 1$ holds.

Proof. In a term of (3.15), multiply the reciprocal by $1 - \theta$. If some stamp has signed dot product $-k$, the normalized term is exactly one. Otherwise its limit is zero. The event that such a stamp exists means that the row pattern opposite the coefficient sign vector occurs in the stamp matrix. Averaging over the 2^k coefficient signs therefore gives the pointwise limit $N_k/2^k$.

There are finitely many canonical column tuples for fixed k, r. Convergence of these normalized kernel values is consequently uniform over that finite set and over all probability mixtures of the tuples. Maximizing over π commutes with this uniform limit. Thus $(1-\theta)G^*_{k,r}(\theta) \to \beta_{k,r}$. Since at least one pattern occurs, $\beta_{k,r} > 0$, and taking reciprocals proves (3.30).

For the lower bound, draw each column uniformly from all 2^r sign columns; this is the uniform canonical distribution followed by independent complements. The resulting r row patterns are independent and uniform on the 2^k possible patterns. Any fixed pattern is absent with probability $(1-2^{-k})^r$. Summing the presence indicators gives the lower bound in (3.31). The upper bound follows because there are at most r distinct patterns and at most 2^k in total. Finally, either repeat the collision argument of Proposition 3.7 at the indicator level, or take its limit as $\theta \uparrow 1$. The resulting bound is $\beta_{k,r} \le 1 - 2^{-r}$ for $k \ge 2$. □

For one stamp, $\beta_{k,1} = 2^{-k}$. For two stamps, an antipodal pair gives $\beta_{k,2} = 2^{1-k}$, recovering the two limits in (3.9). For $k = 2$ and $r = 2, 3, 4, 5$, the coverage payoff is the quadratic quantity β_r in Theorem 3.6. The fixed-family weighted-coloring limit in the next section is complementary: it keeps the support family fixed, whereas (3.30) describes a uniform factor whose extremal family may grow with dimension.

3.9. Realizing the extremal families by nonnegative prices. The constructions above can be embedded in fixed-path packing networks. For (3.6), use k stages in series, each offering two parallel edges with prices $1/k+\theta z_i/k$ and $1/k-\theta z_i/k$. A source indexed by σ uses the edge with its prescribed sign at every stage. Its path price is $1 + (\theta/k)\sigma^T z$. All prices are positive for $\theta < 1$. Add a common monitored edge of price zero and place it on every path. Choosing unit source weights makes its load exactly the sum in the construction. Capacities on the other edges can be chosen large enough not to constrain any of these proposals.

For (3.18), use d stages and add a zero-price bypass at each stage. The source indexed by (S, σ) uses a priced edge, with price $1/k \pm \theta z_i/k$, at precisely the k stages in S, and uses bypasses elsewhere. The common monitored edge is retained. Thus the large-dimensional sharpness examples also use nonnegative component prices and fixed paths. These realizations establish a valid network interpretation of the price families; they do not assert that all their corner configurations are equilibria of one feedback law.

4. Compatibility, stamp thresholds, and residual loss. The reserve factors quantify the worst relative loss from treating all rows independently. They do not tell us how many stamps make the independent envelope attainable on a particular instance. That question depends on signs rather than coefficient magnitudes.

4.1. A graph of incompatible row maxima. The maximum of row s is attained exactly when

$$z_i = -\operatorname{sgn}(b_{si}) \qquad \text{for every } i \in \operatorname{supp}(b_s). \tag{4.1}$$

Coordinates outside the support are unconstrained. Define the sign-conflict graph H on the rows by

$$st \in E(H) \iff b_{si}b_{ti} < 0 \text{ for some } i. \tag{4.2}$$

A constant row is an isolated vertex. A set of row maxima is jointly attainable if and only if the set is independent in H. The forward implication follows from conflicting

requirements on a shared coordinate. For the reverse implication, the absence of any pairwise conflict means that all nonzero requirements on each coordinate have the same sign, so they can be extended to a corner.

THEOREM 4.1 (Exact stamp threshold). *For every nonempty positive family and integer $r \geq 1$,*

$$M_r = R \quad \Longleftrightarrow \quad \chi(H) \leq r. \tag{4.3}$$

Proof. If H has a proper coloring with at most r colors, choose a corner meeting (4.1) for each color class. These corners supply stamps at which every row attains its own maximum. Their load is R, proving $M_r = R$.

Conversely take maximizing copies for M_r. They exist by compactness, and corners suffice by Lemma 2.2. Every row response is at most m_s. If their sum equals $R = \sum_s m_s$, every row must attain its own maximum at some copy. Assign that row to one such copy. Two adjacent rows cannot be assigned to the same copy by (4.1). The assignment is therefore a proper coloring with at most r colors. ∎

The equality is exact; a nearly maximizing row is not considered satisfied. In particular, for $\theta > 0$ the 2^k rows of (3.6) form a clique: any two different sign vectors disagree in at least one coordinate. This family requires 2^k stamps to attain R, even though its two-stamp reserve factor may be close to one for small radius.

The sign graph is not automatically sparse when each row is sparse. Many rows can use a common latent coordinate with opposite signs. Nor does a bounded number of versions imply that the rows in each version form a known independent set. The latter is precisely the additional information that a cohort-specific certificate could exploit.

4.2. Weighted bounds when not all maxima fit. For a nonconstant row define

$$\ell_s = \frac{v_s}{a_s + \rho_s}, \qquad \beta_s^{\min} = \min_{i : b_{si} \neq 0} |b_{si}|, \tag{4.4}$$

$$\delta_s = m_s - \frac{v_s}{a_s - \rho_s + 2\beta_s^{\min}}, \qquad \Delta_s = m_s - \ell_s. \tag{4.5}$$

For constant rows set $\delta_s = \Delta_s = 0$. The value δ_s is a lower bound on the loss of a nonmaximizing row at any corner: at least one minimizing sign is reversed, increasing the denominator by at least $2\beta_s^{\min}$. The value Δ_s bounds any possible loss, including at an interior point.

For a nonnegative vertex-weight vector w let

$$\tau_r(w) = \min \left\{ \sum_{s \in T} w_s : H - T \text{ is } r\text{-colorable} \right\}. \tag{4.6}$$

This is a weighted deletion problem, not a claim that the optimizing set is easy to find.

THEOREM 4.2 (Residual envelope deficit). *For all $r \geq 1$,*

$$\tau_r(\delta) \leq R - M_r \leq \tau_r(\Delta). \tag{4.7}$$

Proof. Choose corner copies attaining M_r, and delete the rows that do not attain their individual maximum at any copy. The remaining rows are r-colorable by the

assignment argument in Theorem 4.1. Each deleted row loses at least δ_s, so the total loss is at least the minimum in (4.6) with weights δ.

For the upper bound, choose a minimizing deletion set for weights Δ. Color the retained graph with at most r colors and choose compatible corners for those color classes. Retained rows then have zero loss. Every deleted row has loss at most Δ_s at its best copy, because its response is at least ℓ_s everywhere. The maximum M_r is at least the value of this constructed set of copies. Rearranging proves the other inequality. ∎

The two bounds have different purposes. The lower bound certifies that a proposed stamp count necessarily leaves a positive gap. The upper bound turns a coloring of a large retained set into a quantitative near-attainment certificate. Both are exact at zero deficit. They can also coincide at positive deficit: the two rows in (1.1) have one conflict edge and, at one stamp, $\delta_s = \Delta_s = 4/3$. Deleting either row gives $R - M_1 = 4/3$.

4.3. A large-radius interpretation. A further limit connects weighted coloring to the size of the loss, not only to its zero set. Fix nonzero vectors d_s with $\|d_s\|_1 = 1$ and consider

$$f_s^{(\theta)}(z) = \frac{v_s}{1 + \theta d_s^T z}, \qquad 0 < \theta < 1. \tag{4.8}$$

The sign graph does not depend on θ. Let $\omega_r(v)$ be the maximum total weight of an induced r-colorable subgraph of H.

PROPOSITION 4.3 (Weighted coloring limit). *For the family* (4.8)*,*

$$\lim_{\theta\uparrow 1}(1-\theta)M_r^{(\theta)} = \omega_r(v), \qquad \lim_{\theta\uparrow 1}\frac{R^{(\theta)}}{M_r^{(\theta)}} = \frac{\sum_s v_s}{\omega_r(v)}. \tag{4.9}$$

Proof. Choose a maximum-weight retained r-colorable set and compatible corners. Every retained row contributes $v_s/(1-\theta)$ and every other contribution is positive. Thus $(1-\theta)M_r^{(\theta)} \geq \omega_r(v)$.

For the converse choose maximizing corners and call a row satisfied if one of these corners meets all its minimizing sign requirements. The satisfied rows induce an r-colorable subgraph, so their total weight is at most $\omega_r(v)$. An unsatisfied row has denominator at least $1 - \theta + 2\theta\eta_s$, where $\eta_s = \min_{i:d_{si}\neq 0} |d_{si}| > 0$. Therefore

$$(1-\theta)M_r^{(\theta)} \leq \omega_r(v) + \sum_s \frac{v_s(1-\theta)}{1-\theta+2\theta\eta_s}.$$

There are finitely many rows, and the last sum tends to zero. This proves the first limit. Since $R^{(\theta)} = (\sum_s v_s)/(1-\theta)$ and $\omega_r(v) > 0$, division gives the second. ∎

This proposition concerns a fixed family followed to the boundary of positive prices. It is different from the large-dimensional sharpness limit in Theorem 3.4, where θ stays fixed. Taking one limit does not justify conclusions in the other regime.

4.4. Support one and the two-stamp collapse. When every row uses at most one latent coordinate, write S_i for the rows depending on coordinate i, and let C_0 be the sum of the constant rows. Then

$$M_1 = C_0 + \sum_i \max_{\epsilon\in\{-1,1\}} \sum_{s\in S_i} \frac{v_s}{a_s + b_{si}\epsilon}. \tag{4.10}$$

The variables separate, so this expression is evaluated using a linear number of rational operations in a sparse input representation. It also follows directly by applying Lemma 2.2 one coordinate at a time.

Two global antipodal corners attain R: whichever sign a nonconstant row needs at its only coordinate appears in one of the two corners. This proves $M_2 = R$ for support one and explains the exact value $K_{1,2} = 1$. The collapse relies on each row having just one sign requirement, not merely on its denominator being scalar-valued.

4.5. Support two can require logarithmically many stamps. Consider the complete-pair family from (3.18), with $k = 2$ and $0 < \theta < 1$. A stamp matrix attains R if and only if, for every pair of columns, its rows contain all four binary sign pairs. This is precisely the strength-two binary covering-array property. Covering arrays and universal test sets are classical combinatorial objects [47, 29, 20]; the connection here is their realization as the saturation threshold of a reciprocal-price envelope.

Let $r_{\min}(d)$ be the minimum number of stamps for this family. Elementary bounds suffice to separate it from the dimension-independent reserve factors:

$$\max\{4, 1 + \lceil \log_2 d \rceil\} \le r_{\min}(d) \le 2\lceil \log_2 d \rceil + 2 \qquad (d \ge 2). \tag{4.11}$$

For the lower bound, four different patterns on any fixed coordinate pair require at least four rows. Moreover, no two columns can be identical or complementary: identical columns omit two patterns, and complementary columns omit the other two. There are at most 2^{r-1} column classes modulo complement, giving $d \le 2^{r-1}$.

For the upper bound, assign the d columns distinct binary codewords of length $L = \lceil \log_2 d \rceil$. Include the all-zero and all-one stamp rows. For each codeword position, include the row of bits in that position and its complement. The first two rows supply patterns 00 and 11 for every coordinate pair. Since two codewords differ in some position, the corresponding row and its complement supply 01 and 10. This gives $2L + 2$ stamps and proves (4.11).

A stronger classical characterization is available. Kleitman and Spencer prove that $r \ge 4$ rows permit at most $\binom{r-1}{\lfloor r/2 \rfloor - 1}$ pairwise qualitatively independent columns, and that this bound is attainable [35]; its use in the modern shattering formulation is also recorded in [2, Section 3]. Therefore the same family has

$$r_{\min}(d) = \min\left\{ r \ge 4 : d \le \binom{r-1}{\lfloor r/2 \rfloor - 1} \right\}.$$

This imported covering-array fact is not needed for our elementary logarithmic bounds and is not claimed as a new extremal result.

The sharp-factor and saturation results can now be compared quantitatively. For fixed $0 < \theta < 1$ and $\varepsilon > 0$, define

$$s_\varepsilon(\theta) = \min\{r \ge 4 : K_{2,r}(\theta) \le 1 + \varepsilon\}.$$

Corollary 4.4 (Approximate recovery versus exact saturation). *As $\varepsilon \downarrow 0$,*

$$\log_2 \frac{\theta(1+\varepsilon)}{\varepsilon} \le s_\varepsilon(\theta) \le \log_2 \frac{1}{\varepsilon} + \frac{1}{2} \log_2 \log_2 \frac{1}{\varepsilon} + O_\theta(1). \tag{4.12}$$

Hence a stamp budget depending only on θ and ε guarantees $R \le (1+\varepsilon)M_r$ for every support-two family, in every dimension. In contrast, exact saturation of the

complete-pair family obeys

$$r_{\min}(d) = \log_2 d + \frac{1}{2}\log_2 \log_2 d + O(1) \qquad (d \to \infty). \tag{4.13}$$

Thus constant-factor approximate recovery is dimension-free, whereas eliminating the final residual loss can require an unbounded number of versions.

Proof. If $K_{2,r}(\theta) \le 1+\varepsilon$, the lower bound in Corollary 3.8 implies $\theta/2^r \le \varepsilon/(1+\varepsilon)$, which is the first inequality in (4.12). Conversely, the upper bound there is at most $1+\varepsilon$ whenever $\theta/(2q_r) \le \varepsilon/(1+\varepsilon)$. Since $q_r = \Theta(2^r/\sqrt{r})$, the right-hand side of (4.12) is sufficient after increasing its $O_\theta(1)$ term. The same central-binomial estimate, inverted in the exact characterization above, gives (4.13). ∎

The approximate statement is uniform over all finite support-two families, while the complete-pair family witnesses the logarithmic exact threshold. Thus a ratio near one does not imply joint attainment of every row maximum, nor does a large saturation threshold imply a large fixed-radius reservation penalty.

4.6. Arbitrary conflict graphs and a saturation dichotomy. Every finite simple graph is a sign-conflict graph at relative radius at most $1/2$. For a graph G of maximum degree $\Delta \ge 1$, orient its edges and use one latent coordinate per edge. The row for vertex v has coefficient $1/(2\Delta)$ on edges leaving v, coefficient $-1/(2\Delta)$ on edges entering v, and zero elsewhere; set $a_v = v_v = 1$. Its radius is $\deg(v)/(2\Delta) \le 1/2$, and two rows conflict exactly when their vertices are adjacent. Isolated vertices use constant rows.

THEOREM 4.5 (Complexity dichotomy for exact saturation). *For every fixed r, deciding $M_r = R$ for rational unrestricted-support instances is polynomial-time solvable when $r \le 2$ and NP-complete when $r \ge 3$. Hardness holds with $a_s = v_s = 1$ and relative radius at most $1/2$.*

Proof. The graph H is computable from coefficient signs. By Theorem 4.1, $r = 1$ asks whether H is edgeless and $r = 2$ whether it is bipartite. For fixed $r \ge 3$, an r-coloring is a witness. Reduce graph three-colorability [25]: for $r = 3$ retain the input graph; for $r > 3$ join it to a clique of size $r-3$. The join is r-colorable exactly when the original graph is three-colorable, because the clique consumes $r-3$ colors and is adjacent to every original vertex. Realize the join by the construction above and apply Theorem 4.1. ∎

This reduction uses one coordinate per incident edge and therefore does not prove fixed-stamp saturation hard under support two. Section 5 instead proves support-two hardness for coherent load maximization.

5. Exact certification and its computational boundary. The sharp reserve factors allow one to relate envelopes without computing their exact values. A controller that wants the full benefit of correlation still needs a sound upper bound on the relevant load. This section establishes both a general obstruction and a structured certificate format.

5.1. A two-coordinate hardness construction. Define the safety decision problem to take a rational positive family and a rational capacity c, and to ask whether $M_1 \le c$. The corresponding unsafe decision is $M_1 > c$. Strictness is important at the boundary, but will not cause an ambiguity in the reduction.

THEOREM 5.1 (Hardness with a fixed positivity margin). *The unsafe decision is NP-complete, and the safety decision is coNP-complete, even when every row has*

exactly two nonzero coefficients, relative radius $1/4$*, and denominator in* $[3,5]$ *over the entire box.*

Proof. Membership of the unsafe decision in NP follows from Lemma 2.2. A corner is a binary witness, and its rational objective value can be evaluated and compared with c in polynomial time in the input encoding length.

Reduce from unweighted Max-Cut, one of the classical NP-complete graph problems [25]. Given a simple graph $G = (V, E)$, use a latent variable z_v for each vertex. For every edge uv add two rows

$$\frac{30}{4 + (z_u - z_v)/2}, \qquad \frac{30}{4 - (z_u - z_v)/2}. \tag{5.1}$$

At a corner, an uncut edge has $z_u = z_v$ and contributes $30/4 + 30/4 = 15$. A cut edge has $z_u - z_v = \pm 2$ and contributes $30/3 + 30/5 = 16$. Hence

$$M_1 = 15|E| + \mathrm{MaxCut}(G). \tag{5.2}$$

This is an exact identity because corners maximize the objective. Each row has coefficient norm one, center four, and denominator between three and five. It suffices to use nonempty graphs and cut targets $1 \le K \le |E|$; inputs outside this range have an immediate answer and can be mapped to fixed yes or no instances. For such an integer target K, set $c = 15|E| + K - 1/2$. Then $M_1 > c$ if and only if G has a cut with at least K edges. The construction and its binary encoding have polynomial size. This proves NP-hardness and the complementary coNP-completeness statement. □

There is also a fixed-path interpretation. Use a common monitored edge with price one. For every vertex make available components with prices $3/2 + z_v/2$ and $3/2 - z_v/2$. The two sources for an edge uv use the positive component for u and negative component for v, or vice versa, together with the common edge. Their path prices are exactly those in (5.1). A staged construction with zero-price bypasses makes these routes fixed paths. Giving each source weight thirty yields the required link load; other capacities can be made large. Thus the reduction does not depend on negative component prices or a price approaching zero.

The support-one formula (4.10) and Theorem 5.1 establish a sharp support boundary for this exact decision problem. They do not rule out useful conservative bounds, approximation algorithms, or efficient algorithms for more structured two-coordinate instances.

5.2. Local factor structure. Make an interaction graph on the latent coordinates by connecting every pair that occurs together in one row. A tree decomposition consists of a tree with bags $B_t \subseteq [d]$ such that every coordinate occurs in a bag, the support of every row is contained in a bag, and the bags containing any fixed coordinate induce a connected subtree. Its width is $w = \max_t |B_t| - 1$.

We take a decomposition as supplied evidence. Finding or approximating one is a separate algorithmic task: fixed-width construction is classical [15], exact exponential algorithms use minimal separators and potential maximal cliques [23], and fixed-parameter construction extends to branch- and rank-decompositions [31]. Recent work gives single-exponential treewidth and branchwidth approximations [36, 22]. Our checker begins after that task. It uses the standard separator property, with reciprocal row values as rational local factors, and verifies a supplied decomposition and local messages rather than certifying the constructor's algorithm.

Root the tree. Let $S_t = B_t \cap B_{\mathrm{parent}(t)}$ be the separator of a nonroot bag, and put $S_0 = \varnothing$ at the root. Assign each row to exactly one bag that contains its support. Write $L_t(\sigma_{B_t})$ for the sum of the rows owned by bag t, evaluated at its corner signs. Define a message by

$$V_t(\sigma_{S_t}) = \max_{\sigma_{B_t \setminus S_t}} \left[L_t(\sigma_{B_t}) + \sum_{u \text{ child of } t} V_u(\sigma_{S_u}) \right]. \tag{5.3}$$

The variables in a child separator are contained in B_t, so every child argument is available inside the maximization. At a leaf the sum of child messages is empty.

THEOREM 5.2 (Exact separator recurrence). *For a valid supplied decomposition and ownership assignment, the root value in* (5.3) *equals M_1. A corner attaining this value can be reconstructed from choices attaining the local maxima.*

Proof. For a bag t, let W_t be the union of bags in its rooted subtree. We prove by induction that $V_t(\sigma_{S_t})$ is the maximum sum of factors owned in that subtree, over assignments to $W_t \setminus S_t$, conditional on the displayed separator signs.

At a leaf this is the definition. For the induction step, the running-intersection property implies that a coordinate appearing in two distinct child subtrees must also appear in the parent bag. Likewise, a coordinate shared by a child subtree and the part of the decomposition above t must appear in its separator. Consequently, after the signs on B_t are fixed, the still-free coordinates in distinct child subtrees are disjoint. Their conditional optima can be attained independently and added. The induction hypothesis gives exactly the bracketed expression in (5.3). Maximizing the remaining bag coordinates proves the conditional interpretation.

The root separator is empty, so the root message maximizes all factors over all corners. Ownership exactly once prevents either omission or double counting. Lemma 2.2 identifies this maximum with M_1. To reconstruct a witness, choose a maximizing root assignment, then descend through maximizing assignments consistent with the already fixed separators. Running intersection ensures that no variable receives incompatible values. ∎

5.3. Upper certificates that do not trust the optimizer. Following the certifying-algorithm principle of separating an untrusted constructor from a small checker [42], a certificate need not assert that any maximization was correctly performed. It may instead provide rational tables U_t satisfying, for every assignment to every bag,

$$U_t(\sigma_{S_t}) \geq L_t(\sigma_{B_t}) + \sum_{u \text{ child of } t} U_u(\sigma_{S_u}). \tag{5.4}$$

The checker verifies the model's positivity, the rooted tree, running intersection, factor ownership, complete separator tables, and all inequalities (5.4). It accepts a capacity claim only if the root table's sole entry is at most that capacity.

THEOREM 5.3 (Soundness and completeness for a supplied decomposition). *Every accepted certificate proves $M_1 \leq U_0(\varnothing) \leq c$. Conversely, if $M_1 \leq c$, the exact messages* (5.3) *form an accepted certificate for any valid supplied decomposition.*

Proof. At a leaf, all inequalities (5.4) imply $U_t \geq V_t$ by taking the maximum over the bag coordinates outside the separator. Suppose this holds for each child. Replacing the child tables on the right side by V_u can only decrease it. Taking the maximum again gives $U_t \geq V_t$. Induction reaches the root, where Theorem 5.2 gives

$V_0 = M_1$. The checked root capacity comparison proves soundness. For completeness choose $U_t = V_t$; a maximum is at least every value it maximizes, so all local inequalities hold, and the root condition follows from $M_1 \le c$. ∎

Completeness is parameterized by the allowed certificate size, which can be exponential in the unrestricted instance. A single bag containing all variables always works, but has 2^d assignments. The coNP-completeness result is therefore consistent with the certificate theorem. A narrow supplied decomposition makes the exhaustive local verification manageable without making the unrestricted problem easy.

For known cohorts, one may certify each cohort's coherent maximum and sum the resulting upper bounds. For unknown cohorts, one may use R, a safe bound for M_r, or a formulation that explicitly represents all copy choices. Copying the latent variables or adding source-to-copy choices can increase the interaction width. The transformed interaction graph requires a new checked decomposition and its own width parameter.

5.4. A four-variable certificate, written out. A small example illustrates why a local upper table is more informative than an optimizer's reported objective. Consider four uncertain coordinates and the load

$$F(z) = 1 + \frac{1}{2 + (z_1 - z_2)/4} + \frac{2}{2 + (z_2 + z_3)/4} + \frac{3}{2 + (-z_3 + z_4)/4}. \tag{5.5}$$

All three nonconstant prices have relative radius $1/4$. The interaction graph is a path. Use bags $B_0 = \{1, 2\}$, $B_1 = \{2, 3\}$, and $B_2 = \{3, 4\}$, in that root-to-leaf order. Assign the constant and first reciprocal to the root, the second reciprocal to B_1, and the third to B_2. The separator signs are respectively empty, z_2, and z_3.

At the leaf, maximizing over z_4 chooses $z_4 = -1$. Its two messages are

$$V_2(-1) = 3/2, \qquad V_2(+1) = 2.$$

At the middle bag, the two options for each fixed z_2 give

$$V_1(-1) = \max\{4/3 + 3/2,\ 1 + 2\} = 3,$$
$$V_1(+1) = \max\{1 + 3/2,\ 4/5 + 2\} = 14/5.$$

Finally, maximizing the root expression over z_1 chooses $z_1 = -1$. The remaining choices give

$$V_0 = \max\{1 + 1/2 + 3,\ 1 + 2/3 + 14/5\} = \max\{9/2, 67/15\} = 9/2.$$

A maximizing global corner is $(-1, -1, +1, -1)$, whose direct load is $9/2$. The rowwise envelope is instead

$$R = 1 + 2/3 + 4/3 + 2 = 5.$$

The difference is not caused by an approximate solver: the middle coordinate requirements prevent all individual maxima from occurring together.

A checker needs neither the preceding maximization choices nor the global maximizing corner. It checks the two leaf inequalities for each separator sign, the middle inequalities using the supplied leaf entries, and the four root inequalities. The numbers $3/2$, 2, 3, $14/5$, and $9/2$ suffice as upper tables. At capacity $9/2$ they establish safety exactly, whereas a purely rowwise calculation would attenuate by $9/10$.

If the claimed root value is reduced while the tables below it are unchanged, the root inequality for $(z_1, z_2) = (-1, -1)$ fails. If only a separate reported objective is

reduced, the checker rejects its disagreement with the root table. Conversely, increasing the root table and its reported bound by one gives a valid but looser certificate; an upper-bound checker must not reject it merely because it is not optimal. These cases distinguish sound majorization, exact optimization, and the separate capacity comparison.

The instance also illustrates the width dependence. There are three bags of size two and only two entries in each nonroot separator table. Replacing the path by a general interaction graph can require much larger bags. The short certificate here is explained by the supplied separator structure, not by the small numerical value of the price radius alone.

5.5. Arithmetic and certificate size. Let B be the number of bags and $L_{\mathrm{scope}} = \sum_s |\operatorname{supp}(b_s)|$ the total factor scope length. Using sparse factor evaluation, construction and verification use

$$O\big((n + L_{\mathrm{scope}} + B)2^{w+1}\big) \tag{5.6}$$

exact rational operations, apart from reading and validating the supplied decomposition. There are at most $B2^{w+1}$ stored table entries. The formula counts each bag assignment, every locally owned factor evaluation, and the child-message accesses. Standard graph searches suffice for the structural checks; these are polynomial in the explicit decomposition size.

For rational inputs, exact message values do not acquire exponentially long binary encodings merely because the maximization enumerates exponentially many assignments. Every exact message is the sum of a subset of the original row responses evaluated at one corner assignment. Each denominator is a rational affine sum of input coefficients. Adding at most n such rational values multiplies denominators in the worst case, so their bit lengths add. The resulting bit length is polynomial in the total input encoding length. Taking a maximum selects one of these rational sums; it does not create a new arithmetic operation that increases precision. Thus (5.6), combined with polynomial-cost integer arithmetic at these bit lengths, gives the usual fixed-width tractability statement.

A maliciously supplied table could contain longer integers than an exact constructor would generate. Verification time must then also depend on the certificate's actual encoding length. Our checker does not infer a precision bound from a claimed treewidth, nor does it use floating-point comparison near capacity.

5.6. The implemented verification contract. The mathematical certificate format and a particular implementation have different acceptance domains. The finite checker used here accepts at most 256 bags, each of size at most 16. These are explicit resource limits on the software, not extra hypotheses in Theorem 5.3. A safe instance whose only supplied decomposition exceeds either limit is rejected by this implementation. Exceeding these limits is inconclusive about safety.

The model and certificate are separate inputs. The model fixes the dimension, positive weights, centers, and coefficient vectors. The certificate supplies bags, parents, owners, separator tables, and a root bound. In particular, the certificate cannot lower a weight or change a coefficient merely by including a different preferred model. The checker recomputes the local reciprocal values from the supplied model and compares exact rational quantities. It does not call the message constructor, its objective evaluator, or its model parser. This is implementation separation, not a claim that two independent research teams or a proof assistant have verified the theorem.

A capacity is also supplied separately when a feasibility decision is required. Acceptance without a capacity establishes only the returned upper bound. Acceptance with a capacity additionally checks that comparison. Neither mode authenticates the origin of the model, the identity of an actuator, or a physical link capacity. A controller must supply the intended model and retain it unchanged while the certificate is checked and used. The mathematical result is conditional on that binding; it does not turn an unauthenticated file into a statement about a real network.

The accepted numerical encodings are integers or exact rational strings. Binary floating-point values and Boolean values are rejected rather than silently interpreted as coefficients. Missing assignments, extra separator assignments, invalid parents, absent variable coverage, and inconsistent ownership are rejected before a capacity is certified. Upward slack in a message is allowed: the certificate's inequalities admit conservative upper tables, so accepted tables can be nonoptimal and nonunique. This accommodates conservative but sound bounds.

Finally, a rejection, interrupted check, or failure to obtain a new certificate does not authorize a new target. Under the serialized command interface, the controller can retain its previously applied feasible command while it recomputes or uses a separately justified conservative envelope. This observation preserves a safety interpretation of verification failure but supplies no progress or availability guarantee. The artifact exercises small trusted synthetic inputs and structural mutations; it is not a hardened parser for unrestricted hostile files.

5.7. A certified rounding option. Exact rational messages can be inconvenient when upper bounds, rather than exact values, suffice. One controlled alternative rounds only upwards. Fix a rational grid spacing $\varepsilon > 0$ and define messages recursively by taking the maximum of the local expression using already rounded child messages, then rounding the result to the smallest multiple of ε no smaller than that maximum.

PROPOSITION 5.4 (Accumulated upward-rounding loss). *The rounded messages form a sound certificate. Their root value exceeds M_1 by at most $B\varepsilon$.*

Proof. The rounded value is at least every unrounded local expression with the rounded children, proving all inequalities (5.4). For the error bound, let $B_t^{\downarrow}$ count the bags in the subtree of t. At a leaf the upward rounding adds less than ε, so the excess over V_t is at most ε. Inductively, for each bag assignment the errors in child messages sum to at most $(B_t^{\downarrow}-1)\varepsilon$. Taking a maximum preserves this bound, because $\max_x a_x - \max_x b_x \le \max_x(a_x - b_x)$. The final rounding contributes at most another ε. Thus the excess at t is at most $B_t^{\downarrow}\varepsilon$, including $B\varepsilon$ at the root. ∎

This proposition is a mathematical option, not a description of the delivered exact checker. A numerical implementation would still need to certify its rounding direction and local comparisons. Ordinary rounding to nearest, followed by an unchecked capacity comparison, is not justified by this argument.

5.8. Trust boundary. The certificate proves an inequality about its fixed input model. It does not prove that an observed price lies in the model, that a source used the claimed stamp, or that a rate command was applied. The model and capacity therefore enter the checker separately from the proposed certificate; the certificate cannot select a more convenient instance for itself. The actuation conditions in the next section supply the additional mathematical steps needed to move from an accepted bound to feasible commanded rates.

6. From a certificate to safe control. An envelope concerns proposed rates at specified price realizations. Safety of a commanded allocation additionally requires that the proposal use the certified information and that changes of allocation respect the packing constraint. We give these steps explicitly, then separate them from convergence of the virtual dual computation. The safety statements concern a discrete command vector in a nonnegative packing polytope. They are conceptually adjacent to positive invariance and control-barrier certification [14, 3], but assume neither plant dynamics nor a differential barrier inequality. Their proofs are order arguments for reciprocal proposals and coordinate applications.

6.1. Largest safe common attenuation. Suppose the proposal family for each link is certified under the actual stamp contract, with exact envelope M_ℓ. Set

$$\gamma_M = \min\left\{1, \min_{\ell: M_\ell > 0} \frac{c_\ell}{M_\ell}\right\}. \tag{6.1}$$

A link with zero envelope contributes no restriction. Every realized proposed rate vector $\widehat{x}$ in the certified family gives a target $x = \gamma_M \widehat{x}$.

PROPOSITION 6.1 (Radial safety and maximality). *Every such target satisfies $Ax \le c$. The value γ_M is the largest common multiplier in $[0,1]$ that works for every proposal in the family. Replacing exact envelopes by certified upper bounds remains safe, but need not give the largest multiplier.*

Proof. For every link, $A_{\ell:}\widehat{x} \le M_\ell$, so $A_{\ell:}x \le \gamma_M M_\ell \le c_\ell$. If $\gamma_M < 1$, at least one positive-envelope link attains the minimum ratio in (6.1). Its envelope maximum is attained by a proposal allowed by the model. Sources not using that link can be assigned arbitrary allowed copies, so this maximizing link proposal extends to a full proposal. Any larger common multiplier makes that link exceed capacity. If $\gamma_M = 1$, no larger multiplier is permitted in the specified interval. The upper-bound version follows from the same chain of inequalities without using attainment. ∎

This is optimality within a one-dimensional family of common multipliers. It is not optimality among all feasible rate allocations. A projection or a new utility maximization may retain more utility at the cost of solving a different problem.

Let γ_R denote (6.1) with the rowwise envelopes R_ℓ. If $M_\ell \le R_\ell \le K M_\ell$ for all used links, then

$$\gamma_R \le \gamma_M \le K\gamma_R. \tag{6.2}$$

Indeed each capacity-to-envelope ratio has this comparison, and clipping the minimum at one preserves it. With $W = \sum_s w_s$, the utility of a positive proposal satisfies

$$U(\gamma\widehat{x}) = U(\widehat{x}) + W\log\gamma. \tag{6.3}$$

Consequently the additional utility loss from using independent rather than the specified correlated envelopes is at most $W\log K$. The applicable K is supplied by the sharp theorems, or by a larger sound per-link factor. This loss comparison uses the same proposal for both filters; it does not compare independently optimized policies.

In the running example, $c = 3$, $M_1 = 8/3$, and $R = M_2 = 4$. One-stamp advice needs no attenuation. A certificate admitting two independent stamps permits the safe common multiplier $3/4$. Treating that delayed proposal as if it had one stamp would incorrectly retain multiplier one.

6.2. Safe asynchronous transitions between targets. Even two feasible targets may have an infeasible coordinatewise mixture. Let $y, z \geq 0$ satisfy $Ay \leq c$ and $Az \leq c$, and put $m_i = \min\{y_i, z_i\}$. Use a serialized epoch with two phases. In the first phase every coordinate is reduced, if necessary, to m_i. The second phase starts only after all these reductions have been acknowledged, and coordinates are then increased, if necessary, to z_i.

THEOREM 6.2 (Decrease-before-increase transition). *Every intermediate commanded vector in this two-phase transition is feasible, regardless of the order of applications within either phase, provided that applications obey their epoch and phase and do not regress to an earlier phase.*

Proof. During the reduction phase, each coordinate lies between m_i and y_i, so the vector is coordinatewise at most y. Nonnegativity of A gives $Ax \leq Ay \leq c$. After all reductions have been applied, the vector is m. During the increase phase each coordinate lies between m_i and z_i, so $Ax \leq Az \leq c$. These statements hold for both atomic changes and monotone continuous interpolation inside a coordinate's allowed interval. ∎

An acknowledgment refers to an applied reduction, not merely receipt of its message. Epoch and phase tags reject stale messages; repeated commands are idempotent; a command from an earlier phase cannot overwrite a later one. No new epoch starts until the preceding target has been reached. These requirements prevent a late reduction or a premature increase from producing a state outside the two bounding boxes used in the proof. If messages may be lost forever or a source may fail before acknowledging, the transition can stop making progress, but the applied state remains in its current feasible phase. Progress requires eventual applied acknowledgments.

Without the phase barrier, apply the target change from $(2, 2/3)$ to $(2/3, 2)$ on a link of capacity three. If the second source increases before the first decreases, the intermediate state is $(2, 2)$ and overloads the link. Both endpoints are feasible. Thus a safe target certificate and a safe transition are logically different requirements.

The construction is a deliberately simple packing-specific actuation rule. It sacrifices simultaneous progress to obtain a short invariant argument. It does not model outstanding packets, queue occupancies, or propagation of traffic already admitted into the network. Those require additional state and service assumptions.

6.3. A restricted virtual dual iteration. We next give sufficient conditions under which the proposal computation itself converges. This is an application of asynchronous contraction reasoning, not a necessary-and-sufficient stability envelope for general primal–dual network dynamics. Feasibility-preserving primal–dual methods also predate the present setting. Bertsekas maintains primal capacity feasibility in a minimum-cost network-flow framework [9], while Awerbuch and Khandekar start from and preserve a feasible packing solution in a distributed positive-LP algorithm [5]. Our safety question concerns a fixed family of bounded reciprocal-price proposals and their stamped mixing, not the first use of feasibility in a primal–dual or distributed packing algorithm.

Let $P = \prod_{i=1}^{m} [\underline{p}_i, \overline{p}_i] \subseteq \mathbb{R}_+^m$ be a compact box on which every path price is positive. Assume it contains a global dual optimum p^* of (2.4). Containing only an optimum of the artificially box-constrained problem is not enough for convergence to the original utility optimum. Define

$$F(p) = \Pi_P\big(p - \operatorname{diag}(\eta_i)\nabla\mathcal{D}(p)\big), \qquad \eta_i > 0, \tag{6.4}$$

where Π_P clips coordinates to their intervals. Because p^* is a global minimizer of $\mathcal{D}$ and lies in P, it satisfies the variational inequality over P and is a fixed point of (6.4).

For each source define its extreme path prices on P by $q_s^{\min} > 0$ and $q_s^{\max}$. The Hessian of the dual is

$$H_{ij}(p) = \sum_s \frac{A_{is}A_{js}w_s}{q_s(p)^2}. \tag{6.5}$$

All entries are nonnegative. The following computable bounds hold throughout P:

$$H_{ii}(p) \geq \mu_i := \sum_s \frac{A_{is}^2 w_s}{(q_s^{\max})^2}, \qquad H_{ij}(p) \leq U_{ij} := \sum_s \frac{A_{is}A_{js}w_s}{(q_s^{\min})^2}. \tag{6.6}$$

Choose positive weights d_i and steps satisfying

$$\mu_i d_i > \sum_{j\neq i} U_{ij} d_j, \qquad 0 < \eta_i \leq U_{ii}^{-1}. \tag{6.7}$$

The first condition implies $U_{ii} > 0$. Define the weighted maximum norm and contraction factor by

$$\|v\|_d = \max_i \frac{|v_i|}{d_i}, \qquad \rho = \max_i \left(1 - \eta_i\mu_i + \eta_i \sum_{j\neq i} U_{ij}\frac{d_j}{d_i}\right) < 1. \tag{6.8}$$

The expression is nonnegative because $\mu_i \leq U_{ii}$ and $\eta_i U_{ii} \leq 1$.

LEMMA 6.3 (Dual contraction). *Under* (6.7), $\|F(p) - F(q)\|_d \leq \rho\|p - q\|_d$ *for all* $p, q \in P$.

Proof. Integrate the Hessian along the segment joining p and q. Its averaged entries retain the bounds (6.6). In row i of the unprojected map's difference, the diagonal coefficient is $1-\eta_i\overline{H}_{ii} \geq 0$, and the off-diagonal coefficients are $-\eta_i\overline{H}_{ij}$. The absolute weighted row sum is therefore at most

$$1 - \eta_i\mu_i + \eta_i \sum_{j\neq i} U_{ij} d_j/d_i \leq \rho.$$

This bounds the unprojected coordinate difference by $\rho d_i\|p-q\|_d$. Scalar interval projection is nonexpansive, so projecting that coordinate cannot increase the difference. Divide by d_i and take the maximum. ∎

The diagonal-dominance condition can fail on a perfectly valid network, for example when shared-path coupling dominates the diagonal bounds. Such failure is not a divergence theorem. It only places that instance outside the contraction result just proved.

6.4. Asynchronous reads, update gaps, and errors. At slot $t \geq 0$, each updated coordinate i forms a mixed read $\widetilde{p}^{i,t} \in P$, whose coordinate j comes from some state time between $t - D$ and t. Histories at times $-D, \ldots, 0$ are supplied in P. The update is

$$p_i^{t+1} = \Pi_{P_i}\left(\widetilde{p}_i^{i,t} - \eta_i \nabla_i \mathcal{D}(\widetilde{p}^{i,t}) + e_i^t\right). \tag{6.9}$$

Coordinates not updated retain their current value. Several coordinates may update at one slot, using different mixed reads. Assume that every coordinate is updated in

each interval of B consecutive slots, and that $|e_i^t|/d_i \le \varepsilon$ whenever it updates. Thus D bounds read age and B bounds the update gap. Neither assumption follows just from the word asynchronous.

THEOREM 6.4 (Bounded-delay error envelope). *Put $T = D + B$ and define*

$$E_0 = \max_{-D \le t \le 0} \|p^t - p^*\|_d, \qquad R_\varepsilon = \frac{\varepsilon}{1-\rho}, \qquad \overline{E} = \max\{E_0, R_\varepsilon\}.$$

Then for every $t \ge 0$,

$$\|p^t - p^*\|_d \le R_\varepsilon + \rho^{\lfloor t/T \rfloor}(\overline{E} - R_\varepsilon). \tag{6.10}$$

If the additive errors tend uniformly to zero across updates, the virtual states converge to p^*.

Proof. An updated coordinate obeys

$$\frac{|p_i^{t+1} - p_i^*|}{d_i} \le \rho \|\widetilde{p}^{i,t} - p^*\|_d + \varepsilon.$$

This follows from the unprojected row estimate in Lemma 6.3, interval nonexpansiveness, and the additive-error bound. If every available historical coordinate is within $\overline{E}$, the new coordinate is within $\rho\overline{E} + \varepsilon \le \overline{E}$. Unchanged coordinates retain that bound. Induction gives a uniform bound $\overline{E}$ for all times.

Put $E_j = R_\varepsilon + \rho^j(\overline{E} - R_\varepsilon)$. We prove the stronger tail statement that every state at time $t \ge jT$ is within E_j. The case $j = 0$ was just proved. Suppose the statement holds for $j-1$. Every update in a slot $u \ge (j-1)T + D$ reads only coordinates whose state times are at least $(j-1)T$, so its new value is within $\rho E_{j-1} + \varepsilon = E_j$. Within the next B slots every coordinate has such an update. At state time $(j-1)T + D + B = jT$ all coordinates therefore satisfy the new bound. All subsequent updates still read times at least $(j-1)T$ and preserve it. This proves the tail statement and hence (6.10).

For vanishing errors, fix any $\varepsilon' > 0$. After some finite time all updates have normalized error at most ε'. Restart the preceding argument with the then-current bounded history, which lies in the compact box P. It gives $\limsup_t \|p^t - p^*\|_d \le \varepsilon'/(1-\rho)$. Since ε' is arbitrary, the limsup is zero. ∎

This proof handles mixed reads rather than silently replacing them with one coherent past vector. Its epoch length includes both purging old reads and covering all coordinates with updates. For a fixed persistent error, it proves a neighborhood bound, not convergence to the original optimizer.

Example 6.5 (A safe biased limit). Take one source and one link with $w = c = 1$. The unprojected dual update with step $\eta = 1/2$ and constant additive error $e = 1/8$ is

$$p^+ = p - \frac{1}{2}(1 - 1/p) + \frac{1}{8}.$$

At $p = 4/3$ the change is zero, while the inverse-price rate is $3/4$. Choose a box containing both 1 and $4/3$ and separated from zero so that the fixed point is not clipped. The resulting allocation is feasible but not optimal; the true optimum has rate one and price one. Thus bounded advice error alone cannot imply exact optimality, even without delay or multiple sources.

6.5. A nontrivial instance satisfying the contraction condition. The hypotheses are restrictive but not empty beyond scalar control. Consider

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad w = (4, 4, 1), \quad c = (5, 5), \quad P = [4/5, 1]^2. \tag{6.11}$$

The price $p^* = (9/10, 9/10)$ gives rates $(40/9, 40/9, 5/9)$, and each link carries exactly five. Stationarity, primal feasibility, and complementary slackness establish the global optimum. On the stated box,

$$\mu_1 = \mu_2 = 17/4, \qquad U_{11} = U_{22} = 425/64, \qquad U_{12} = U_{21} = 25/64.$$

With $d_1 = d_2 = 1$ and $\eta_1 = \eta_2 = 1/8$, the steps meet (6.7) and $\rho = 265/512$. Thus the error-free virtual iteration contracts by at least this factor per $D + B$ slots under the schedule assumptions. These rational calculations are a worked analytic instance, not measured networking performance or a claim that this step size is optimal.

6.6. A rate bound when error tolerances decrease. The preceding convergence statement can be made quantitative without replacing decreasing errors by their largest value forever. This is useful when exact rational computations are quantized at a progressively finer grid.

Suppose ϵ_t is nonnegative and nonincreasing and every update in slot t satisfies $|e_i^t|/d_i \le \epsilon_t$. Choose

$$E_0 \ge \max\left\{ \max_{-D \le t \le 0} \|p^t - p^*\|_d, \frac{\epsilon_0}{1-\rho} \right\},$$

and, with $T = D + B$, define

$$E_j = \rho E_{j-1} + \epsilon_{(j-1)T}, \qquad j \ge 1. \tag{6.12}$$

PROPOSITION 6.6 (Decreasing-error tail levels). *The sequence E_j is nonincreasing, and every state at time $t \ge jT$ satisfies $\|p^t - p^*\|_d \le E_j$. Moreover,*

$$E_j = \rho^j E_0 + \sum_{h=1}^{j} \rho^{j-h} \epsilon_{(h-1)T}. \tag{6.13}$$

If $\epsilon_t \to 0$, then $E_j \to 0$.

Proof. The choice of E_0 gives $E_1 \le E_0$. If $E_j \le E_{j-1}$, monotonicity of ϵ_t yields

$$E_{j+1} = \rho E_j + \epsilon_{jT} \le \rho E_{j-1} + \epsilon_{(j-1)T} = E_j.$$

Thus the levels decrease. The uniform initial bound follows as in Theorem 6.4, using $\epsilon_t \le \epsilon_0$.

Suppose all state times at least $(j-1)T$ are within E_{j-1}. Every update at a slot $u \ge (j-1)T + D$ reads only those state times and has error at most $\epsilon_{(j-1)T}$. Its output is therefore within E_j. The update-gap assumption ensures that all coordinates have such an output by state time jT. Subsequent outputs preserve the same bound. This proves the tail assertion by induction.

Unrolling the scalar recurrence gives (6.13). For its limit, fix $\delta > 0$ and split the sum into finitely many early terms and the remaining terms where $\epsilon_{(h-1)T} \le \delta$. Each early term tends to zero as j grows, while the remaining geometric sum is at most $\delta/(1-\rho)$. Letting δ tend to zero proves the claim. ∎

This bound applies to prescribed error tolerances; it does not infer prediction accuracy from the observed iterates. For instance, rounding an unprojected scalar update down to a grid of spacing h_t creates additive error in $(-h_t, 0]$. Clipping the rounded value to P_i preserves the error analysis because the theorem places the perturbation before the same interval projection. If the norm weights are unequal, the normalized error bound is h_t/d_i, not simply h_t. Rounding after an uncontrolled nonlinear operation would require its own error accounting.

6.7. A complete interval-certified command interface. A simple interface can certify all finitely delayed prices without assuming that different links' uncertainties are favorably correlated. It is conservative, but closes the logical chain from virtual iterates to commanded rates. At a command epoch, let $\mathcal{H}$ be the finite set of price vectors that any source is allowed to use. It includes every retained version that can still produce a proposal in that epoch. Define coordinate bounds

$$\ell_i = \min_{p\in\mathcal{H}} p_i, \qquad u_i = \max_{p\in\mathcal{H}} p_i.$$

All vectors lie in P, so the interval box $[\ell, u]$ does too. Since $A \geq 0$, every path price is minimized at the same lower corner. Therefore the exact envelope under this box, for any stamp count, is

$$\overline{x}_s = \frac{w_s}{\sum_i A_{is}\ell_i}, \qquad M_a = \sum_s A_{as}\overline{x}_s. \tag{6.14}$$

The index a here denotes a monitored link. Every denominator is positive by the global price-box assumption. Computing (6.14) uses a number of arithmetic operations linear in the sparse routing description after the coordinate extrema are known.

A source chooses any one of its allowed versions, producing $\widehat{x}_s = w_s/q_s(p^{\tau_s})$. Different sources may choose different versions. Compute γ from (6.1), certify the target $x = \gamma\widehat{x}$, and apply the transition of Theorem 6.2. The price enclosure and the set of allowable source versions are frozen for that command epoch. Subsequent virtual updates may continue, but a newer version cannot be substituted into the already certified proposal. It belongs to a later command epoch with a new enclosure.

This construction has $M_1 = M_r = R$ for its interval model. It is therefore an explicit baseline rather than evidence of the correlation benefit proved earlier. A certifier with a justified signed affine description can replace (6.14) by the tighter envelope developed in Sections 3–5, but must still enclose every permitted source version and preserve the same epoch rule.

THEOREM 6.7 (Quantitative rate and utility transfer). *Let $x^* = w/(A^T p^*)$ be the utility-optimal rate vector. Suppose every vector in $\mathcal{H}$ obeys $\|p - p^*\|_d \leq E$. Define*

$$C_s = \sum_i A_{is} d_i, \quad L_s = \frac{w_s C_s}{q_s^* q_s^{\min}}, \quad \kappa_s = \frac{C_s}{q_s^*}, \quad \Lambda = \max_a \frac{\sum_s A_{as} L_s}{c_a}. \tag{6.15}$$

Then the interval-certified target is feasible and satisfies

$$\gamma \geq (1 + \Lambda E)^{-1}, \tag{6.16}$$

$$|x_s - x_s^*| \leq (L_s + \Lambda x_s^*)E, \tag{6.17}$$

$$\frac{x_s}{x_s^*} \geq \frac{1}{(1+\Lambda E)(1+\kappa_s E)}, \tag{6.18}$$

$$0 \leq U(x^*) - U(x) \leq W\log(1+\Lambda E) + \sum_s w_s \log(1+\kappa_s E). \tag{6.19}$$

For an intermediate state between two such targets, the coordinate-ratio and utility bounds hold with E replaced by the larger of the two endpoint error bounds.

Proof. Every coordinate of the lower corner ℓ is still within $d_i E$ of p_i^*. For either an allowed price vector or ℓ, the path-price discrepancy is at most $C_s E$. Both its denominator and q_s^* are positive, with the former at least $q_s^{\min}$. The reciprocal identity consequently gives

$$|w_s/q_s - x_s^*| \le L_s E.$$

In particular, $M_a \le (Ax^*)_a + E\sum_s A_{as}L_s \le c_a(1+\Lambda E)$. Taking capacity ratios and clipping at one proves (6.16). Feasibility follows from Proposition 6.1.

For a realized proposal, write

$$|\gamma\widehat{x}_s - x_s^*| \le \gamma|\widehat{x}_s - x_s^*| + (1-\gamma)x_s^*.$$

Use the reciprocal bound, $\gamma \le 1$, and $1-\gamma \le \Lambda E/(1+\Lambda E) \le \Lambda E$ to obtain (6.17). Also $q_s(p^{\tau_s}) \le q_s^* + C_s E$, so

$$\widehat{x}_s/x_s^* = q_s^*/q_s(p^{\tau_s}) \ge (1+\kappa_s E)^{-1}.$$

Together with (6.16), this proves (6.18). Taking logarithms, multiplying by w_s, and summing gives the upper utility bound. Its nonnegativity follows from feasibility of x and optimality of x^*, not from a componentwise ordering of their rates.

Every intermediate coordinate of the two-phase transition is at least the smaller of its endpoint coordinates. The right side of (6.18) is nonincreasing in E. Thus both endpoints, and hence the intermediate coordinate, satisfy that lower bound with the larger error. Intermediate feasibility follows from Theorem 6.2; the same logarithmic argument completes the proof. ∎

When $\mathcal{H}$ consists of versions with state times in $[t-D, t]$, the tail bounds give an explicit choice of E. For $t \ge D$, one may use the level with index $\lfloor (t-D)/T \rfloor$ in Proposition 6.6; for earlier epochs use the initial uniform level. The resulting error bound depends on retained versions, not only on the newest iterate. Because $\log(1+v) \le v$ for $v \ge 0$, the utility loss also admits the linear upper bound $E(W\Lambda + \sum_s w_s\kappa_s)$. This worst-case certificate-to-utility transfer bounds utility loss from a prescribed error envelope; the optimal convergence rate is undetermined.

In the two-link example (6.11), the constants are

$$L = (50/9, 50/9, 25/36), \qquad \kappa = (10/9, 10/9, 10/9), \qquad \Lambda = 5/4.$$

All are rational. They permit a finite test to verify the rate inequalities directly without approximating logarithms: it suffices to check the coordinate lower bound (6.18). The separate logarithmic step is supplied by the proof. The interface remains valid if the virtual process fails to converge, provided the epoch enclosure and phase conditions continue to hold; only the shrinking bounds on performance are then lost.

6.8. When safety conservatism vanishes. Convergence of the virtual state does not imply that a fixed nonzero uncertainty box shrinks. To obtain convergence of the filtered targets to the utility optimum, suppose the certified price families at epoch t satisfy

$$a_s^t \longrightarrow q_s^* > 0, \qquad \|b_s^t\|_1 \longrightarrow 0, \tag{6.20}$$

where $q^* = A^T p^*$. This condition must enclose all retained stamps, not merely the newest advice. It can hold, for example, when all finitely delayed reads of a convergent

price process approach the same limit and the certified additional advice error also vanishes.

PROPOSITION 6.8 (Uniform disappearance of filtering). *Under* (6.20)*, every admissible proposed rate converges uniformly to* $x^*_s = w_s/q^*_s$*. Every link envelope* M^t_ℓ*, for any allowed number of stamps, converges to* $(Ax^*)_\ell$*. The exact multiplier* (6.1) *tends to one, and its filtered targets tend to* x^**. Applying the serialized two-phase transitions preserves this limit provided each epoch completes in finite time and the completed epoch index tends to infinity.*

Proof. Let $e^t_s = |a^t_s - q^*_s| + \|b^t_s\|_1$. For all sufficiently large t, $e^t_s < q^*_s$. Every certified denominator is within e^t_s of q^*_s, so

$$\left| \frac{w_s}{a^t_s + (b^t_s)^T z} - \frac{w_s}{q^*_s} \right| \le \frac{w_s e^t_s}{q^*_s(q^*_s - e^t_s)}. \tag{6.21}$$

This tends to zero uniformly in z. Choosing different stamps for different rows cannot enlarge a bound that already holds for each response individually. Multiplying by routing coefficients and summing over finitely many sources proves uniform convergence of every possible link load and hence of its maximum.

If the limiting link load is positive, c_ℓ/M^t_ℓ tends to a value at least one, because $Ax^* \le c$. A link with zero limiting load has capacity strictly positive and eventually imposes no restriction. Taking the finite minimum and clipping at one gives $\gamma^t_M \to 1$. The filtered targets therefore converge to x^*.

Every intermediate coordinate during a transition lies between the corresponding coordinates of its two endpoint targets. Its distance from x^*_i is at most the larger endpoint distance. Once both consecutive endpoint targets are close to x^*, the entire transition is close. Finite completion and an unbounded completed epoch index exclude remaining forever in an early transition, proving the final assertion. ∎

If certificates use upper bounds $U^t_\ell \ge M^t_\ell$, nonvanishing slack can keep the multiplier obtained from (6.1) with U^t below one even under (6.20). Conversely, the filtering-disappearance conclusion of Proposition 6.8 extends to these bounds when (6.20) holds and $U^t_\ell - M^t_\ell \to 0$ for every link: then $U^t_\ell \to (Ax^*)_\ell$, so the multiplier tends to one and the filtered targets tend to x^*. For one link with $c = M^* = 1$, $U^t = 1 + 1/t$ gives $t/(t+1) \to 1$, whereas fixed $U^t = 2$ gives $1/2$.

7. Exact finite evidence. The mathematical claims above concern arbitrary finite families. Exact rational computation is used to attack identities, quantifier order, implementations, and boundary cases, not to replace proof. No predictor is trained and no packet network is simulated.

7.1. Checked obligations. Table 1 summarizes the retained cases. Independent oracles either assign rows to stamps before corner enumeration or choose a corner tuple before taking each row's best stamp; agreement checks the order of maxima in M_r.

Closed forms are checked at radii $1/4$, $1/2$, and $3/4$. Complete support-two families use exhaustive multiplicity enumeration for $r \in \{2,3,4\}$ and $2 \le d \le 10$, and for $r = 5$ and $2 \le d \le 7$. Direct tuple oracles cover $d \in \{2,3\}$ and $r \in \{2,3,4,5\}$. The five-stamp check independently verifies the split classification, the incidence identities for $L(K_5)$, the multiplicities of its three eigenvalues by exact ranks, and the formula for $P_5(d)$ on explicit constructions through $d = 40$.

The hardness test covers all 1,099 simple graphs on at most five vertices, comparing an independent Max-Cut routine with (5.2). Independent coloring, stamp,

TABLE 1
Executed exact checks. Counts are cases or comparisons, not general proofs.

Obligation	Count
Sign-family extremizers / zero-radius cases	30/6
General one-/two-stamp inequalities	240/48
Palette evaluations / inequalities / sampling comparisons	9/12/6
Variational grids / all-stamp support-two sandwiches	22/15
Two-coordinate extrema / inequalities / direct oracles	33/72/8
Five-stamp spectral certificate / explicit constructions	1/39
Max-Cut reduction graphs	1,099
Coloring / fixed-r reductions / deletion / copy cases	285/225/285/9
Covering constructions / exact minima	15/3
Certificate oracles / rejected mutations / boundary cases	72/20/3
Transition orders / generic contraction schedules	7,404/4
Two-link runs / slots / intermediate states	12/768/4,608
Correlated stamp tuples / epochs / intermediate states	512/8/65,536

deletion, and covering checks are supplemented by 225 join-reduction cases for $r \in \{3, 4, 5\}$, including conflict-graph identity, radius at most $1/2$, and the required coloring equivalence.

7.2. Negative controls. The scalar pair in Section 2 has $M_1 = 8/3$ but $M_2 = R = 4$, so capacity three is coherent-safe and mixed-unsafe. At $\theta = 1/2$, the support-two sign family gives

$$R = 8, \qquad M_1 = 14/3, \qquad R/M_1 = 12/7 > 3/2 = K_{1,1}(1/2),$$

refuting substitution of the scalar factor. The persistent-error example in Section 6 stays feasible but converges to the wrong optimum. Seventy-two deterministic path-decomposition cases compare the constructor and checker with direct full-cube enumeration; each exact capacity is accepted and its value minus $1/100$ rejected. Twenty certificate mutations and three boundary cases test malformed messages, structure, ownership, encodings, bounds, degeneracy, and loose-but-sound tables.

7.3. Magnitude of the factors. Figure 1 plots theorem values, not empirical fits. The one- and two-stamp constants are attained by a finite sign family; the three-, four-, and five-stamp values are suprema approached by growing complete support-two families. At radius $1/2$ the five factors are $12/7$, $4/3$, $6/5$, $12/11$, and $16/15$. A finite family may have a smaller ratio: in dimension four the complete support-two family has three-, four-, and five-stamp ratios $48/41$, $24/23$, and 1.

7.4. Certified variational intervals. For $k = 2$, $\theta = 1/2$, and $r \le 5$, twenty-two rational grids enumerate all selected weak compositions. At two stamps and denominators $Q = 1, 4, 16, 64$, the certified $K_{2,2}$ intervals are $[1, 4/3]$, $[1, 4/3]$, $[6/5, 4/3]$, and $[48/37, 4/3]$. Every closed form lies in its interval; these are fixed-parameter checks, not large-r performance evidence.

7.5. Correlation-aware command experiment. Use the complete support-two family in dimension three at $\theta = 1/2$: twelve unit-weight sources share one link. Enumeration of all $8^3 = 512$ corner triples gives $M_3 = 21 < 24 = R$. At capacity 21,

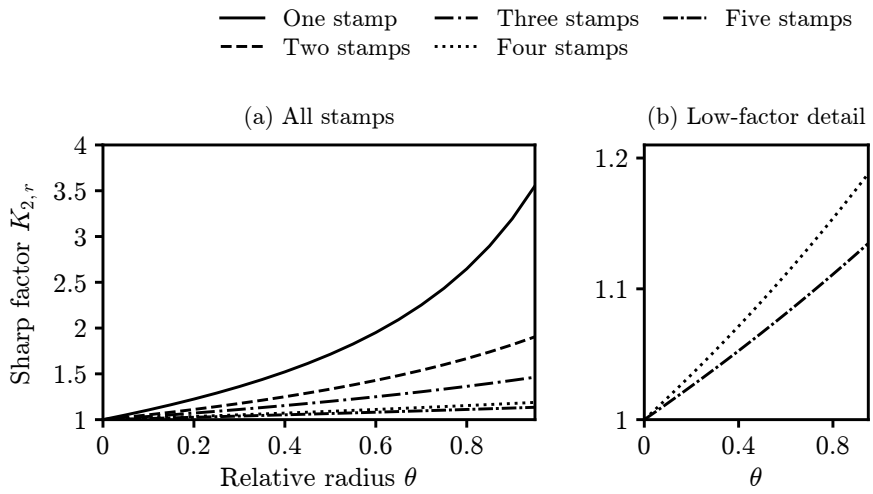


FIG. 1. *Sharp support-two factors from one through five stamps. Panel (b) repeats the four- and five-stamp curves on an expanded vertical scale. The fifth curve is* $K_{2,5} = 8/(8-\theta)$.

the correlation-aware attenuation is one, while the rowwise envelope permits $7/8$, an exact proportional-utility loss of $12\log(8/7)$ on the same proposal.

Across eight cyclic Gray-code epochs, every subset of completed decreases and every subset of completed increases is checked, totaling 65,536 intermediate states; all loads are at most 21. Componentwise mixing of one target with its global complement instead sets all twelve rates to two and has load $24 > 21$. This tests both retained correlation and ordered actuation, not traffic performance.

7.6. Delayed virtual-state experiment. A separate exact process runs (6.11) for 64 slots under $D \in \{0, 1, 2\}$, two starts, and fixed or refining quantization. The retained epoch box is recomputed after every update, and all decreases are acknowledged before increases. All 768 targets and 4,608 intermediate states satisfy both capacities and Theorem 6.7's lower bound.

For the $D = 2$ trajectory in Figure 2, the slot-64 errors are $7/20480$ and $3/1310720$ for fixed and refining grids. This sign-compatible case has $M_1 = M_r = R$ and isolates delayed contraction; the preceding experiment tests the correlation gain.

8. Consequences and limits. Intervals discard joint structure; one stamp retains a shared realization; r stamps retain it only within versions. Reserve factors price this loss, and the conflict graph characterizes when versions realize all rowwise maxima.

8.1. Interpretation. A bound on R/M_r measures conservatism, not the utility of an arbitrary controller. The logarithmic-utility comparison applies when the same positive proposal is filtered by two common attenuations. Nor does a ratio near one imply a small saturation threshold: complete support-two families need $\Theta(\log d)$ stamps even though Corollaries 3.8 and 4.4 make $(1+\varepsilon)$-approximate recovery dimension-free.

For fixed k, r, the column-distribution polynomial is exact, not uniformly efficient; numerical solutions outside proved closed forms need certified upper bounds. Coloring, covering arrays, decomposition, separator dynamic programming, and column frequencies are existing tools. The new objects are the reciprocal partial-coverage

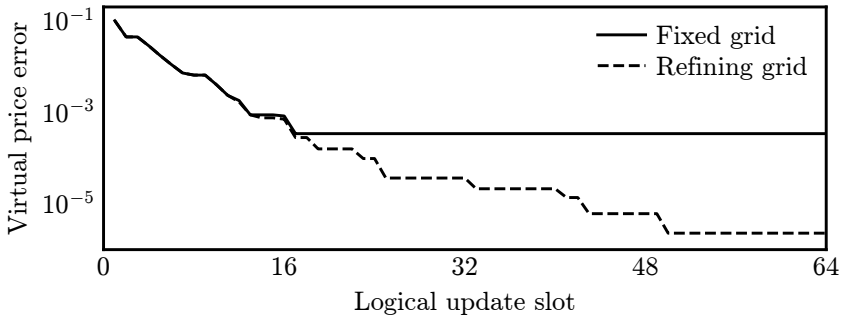


FIG. 2. *Exact two-link experiment with $D = 2$, initial price $(4/5, 1)$, alternating updates, and fixed or refining quantization. The curve is virtual-price error, not packet-network performance.*

payoff, its symmetrization, and the resulting information contract.

8.2. Boundaries. A certificate applies to one fixed affine family with positive denominators. Moving its center, discarding retained versions, or changing clipping rules requires showing that every active value remains enclosed. Delay can bound how many versions are retained, but alone supplies neither statistical independence nor a smaller reachable set. Independent nonnegative link-price intervals are sign compatible and may exhibit no correlation advantage; signed latent dependence is substantive.

The five-stamp $L(K_5)$ spectrum is support-two specific; larger stamp counts introduce other split sizes, so Corollary 3.8 remains the general statement. The one-link experiment isolates actuation, while routed networks require consistent per-link stamp semantics.

The transition theorem concerns commanded rates in a nonnegative packing system. It assumes phase tags, applied acknowledgments, and rejection of stale commands; lost acknowledgments may stop progress. Packets in flight, queue dynamics, link failures, and Byzantine sources are outside the model. Likewise, virtual-price convergence requires a global dual optimum in the chosen box, weighted diagonal dominance, bounded read ages and update gaps, and the stated error regime. Safety may remain valid when those convergence hypotheses fail.

8.3. Conclusion. For positive reciprocal-affine responses, contemporaneous correlation has an exactly bounded value. Sharp formulas, an all-stamp support-two sandwich, a coloring characterization with a polynomial/NP-complete saturation dichotomy, and a coNP-complete certification boundary separate statistical description, combinatorial compatibility, and computation. Separator messages and a decrease-before-increase interface then turn sound envelope bounds into checkable commanded-rate safety. Exact convergence and disappearance of conservatism require the additional dynamical and shrinking-uncertainty assumptions stated in Section 6.

Code Availability. Code and accompanying artifact materials are available at https://github.com/Haoyi-Zhang/PriceEnvelope.

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